Customer Needs and User Research

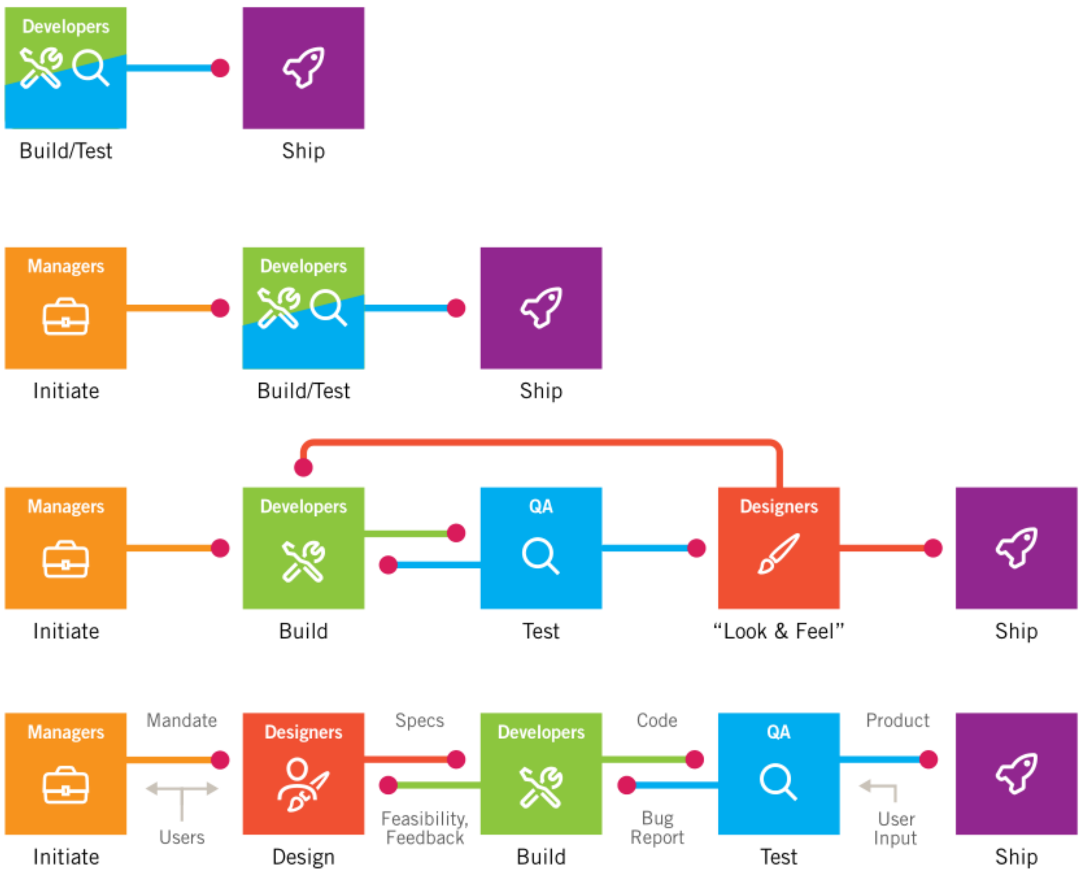
Lecture 3

Outline

- Product Development Process
- Researching Users
- Identifying Customer Needs
- Engineering Idea #2: Ant Colony

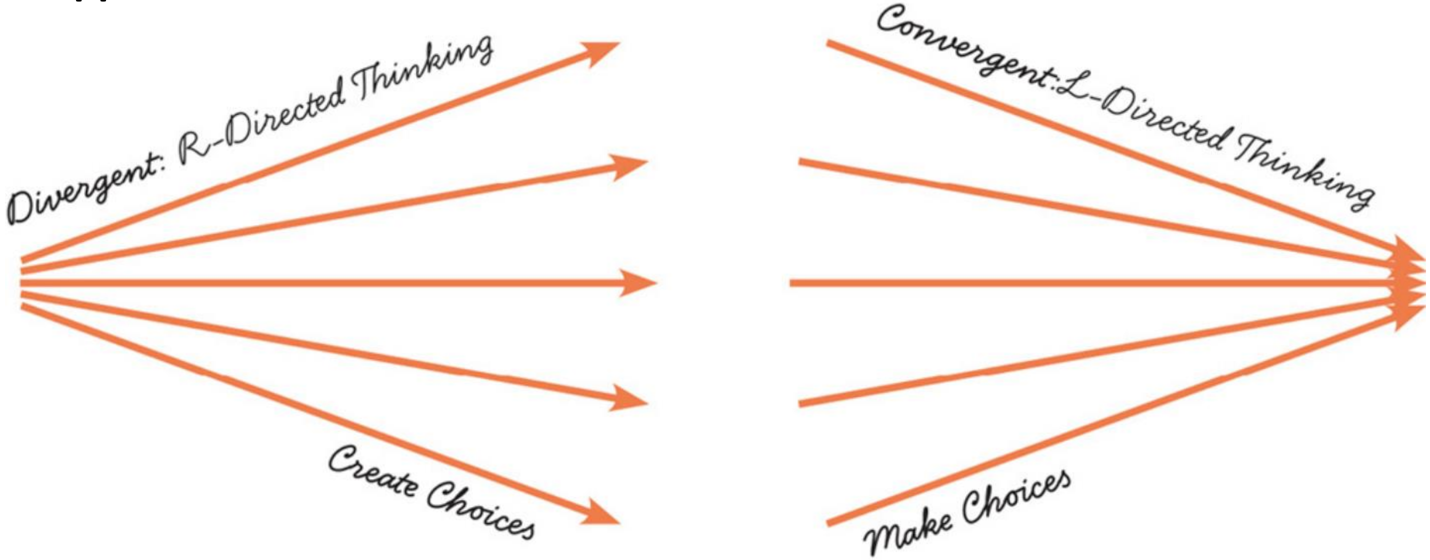
Product Development Process

How Product Development Evolves?

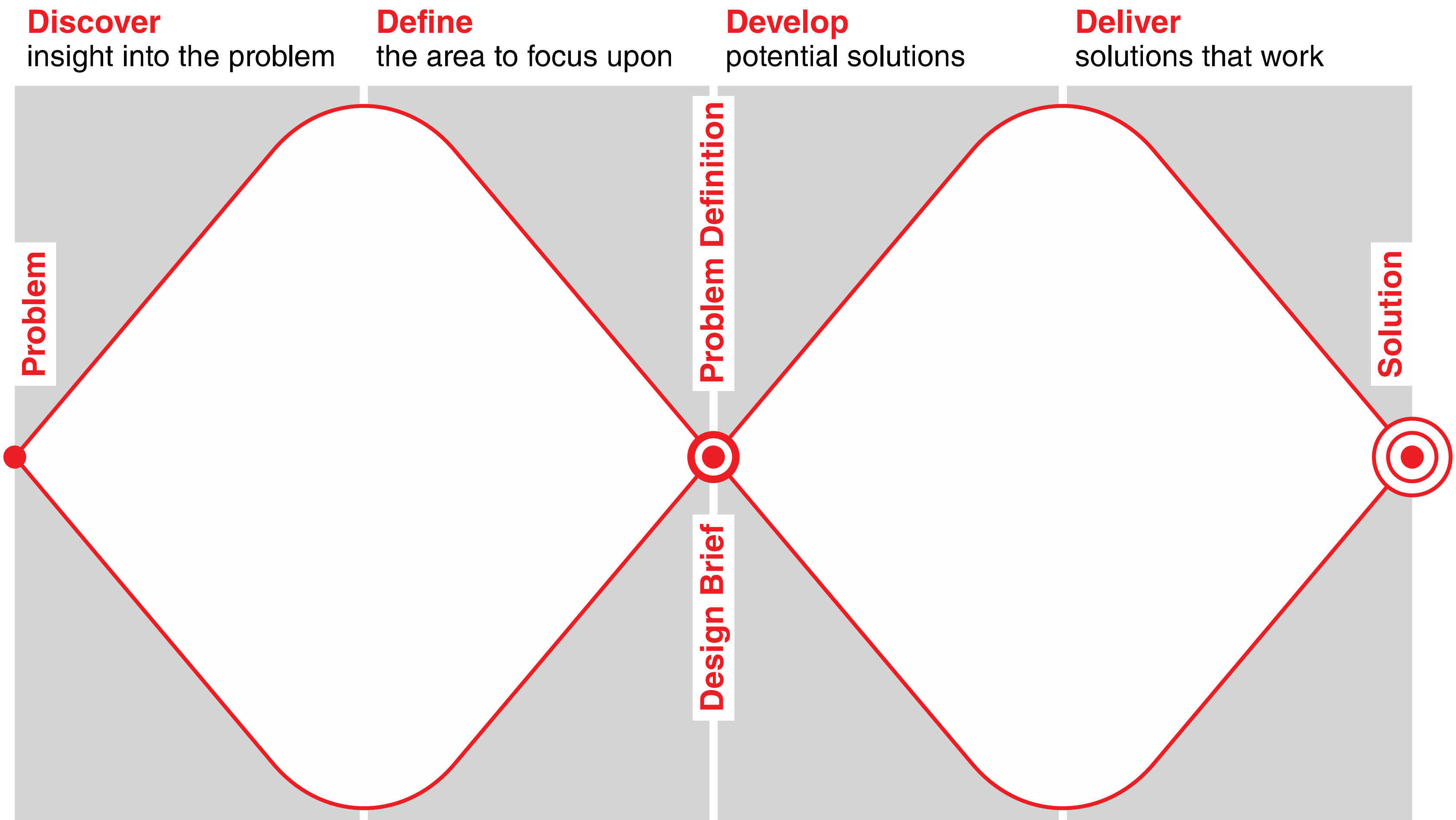


Divergent and Convergent Thinking

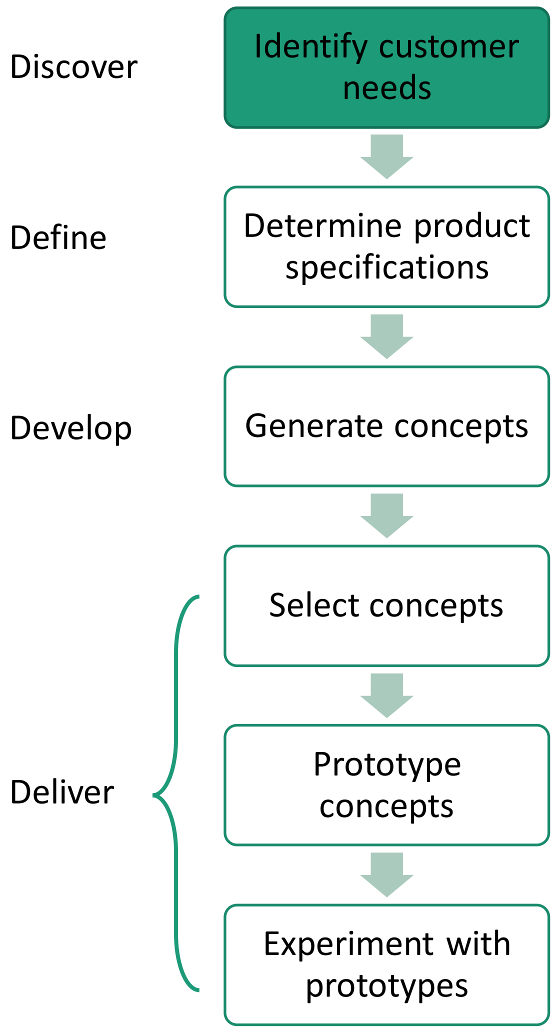
- Divergent Thinking
  - Ideas and thoughts are followed and explored in many directions
  - Ideas are generated and recorded to look back when needed.
  - Associated with **creativity, randomness, visualizations, emotions, extraordinary** and **non-judgemental approaches** etc.
- Convergent Thinking
  - Ideas and thoughts are focused on finding a single correct answer to a problem
  - Ideas are generated, weighed and discarded until the correct solution is found.
  - Associated with **logic, analysis, scientific scrutiny, conventional** and **systematic approaches**.



Design Thinking Process

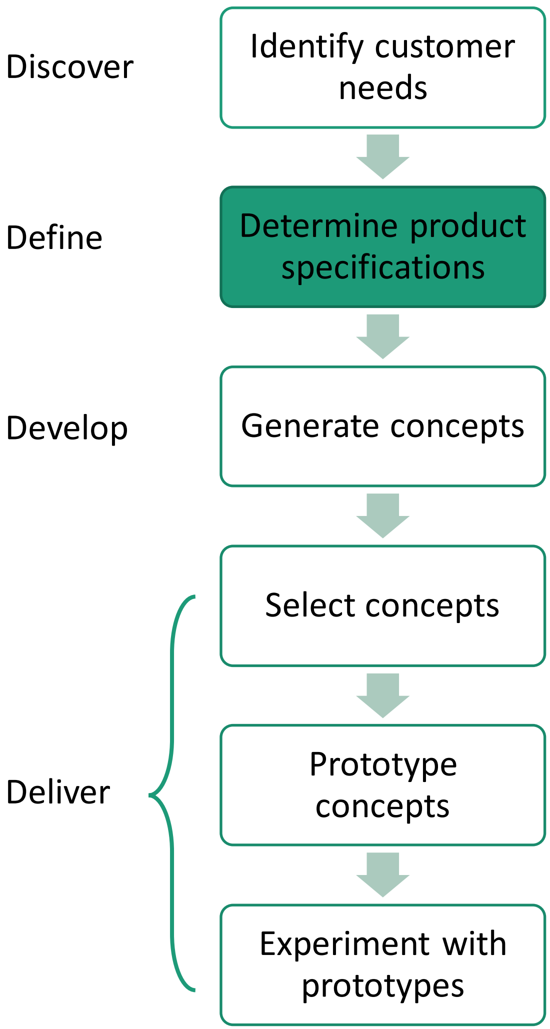


Systematic Design Procedure



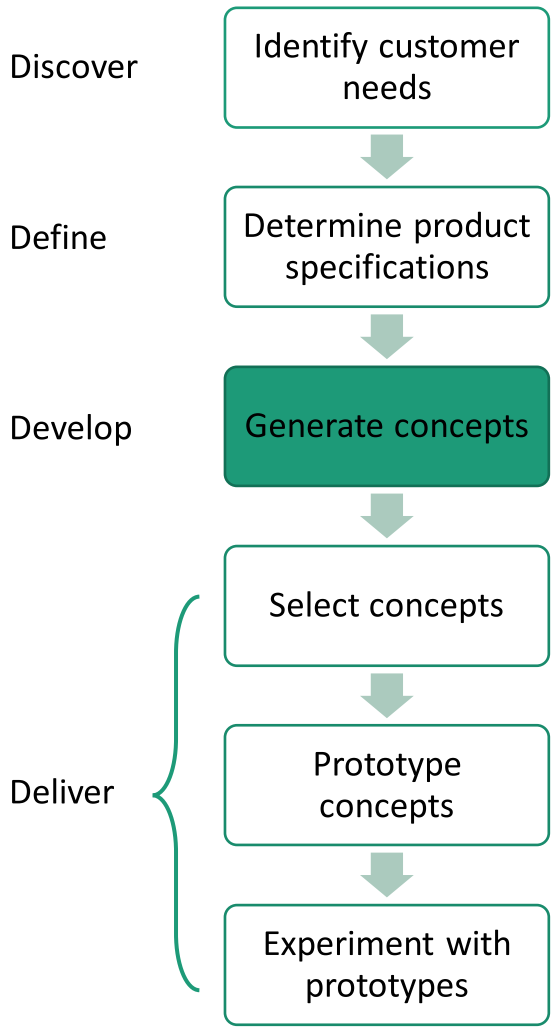
- Analyze market and identify customer needs
- What would be the feature of the product
- Benchmarking of other products
- Consider constraints including legal regulations, industrial standard, and environmental regulations

Systematic Design Procedure



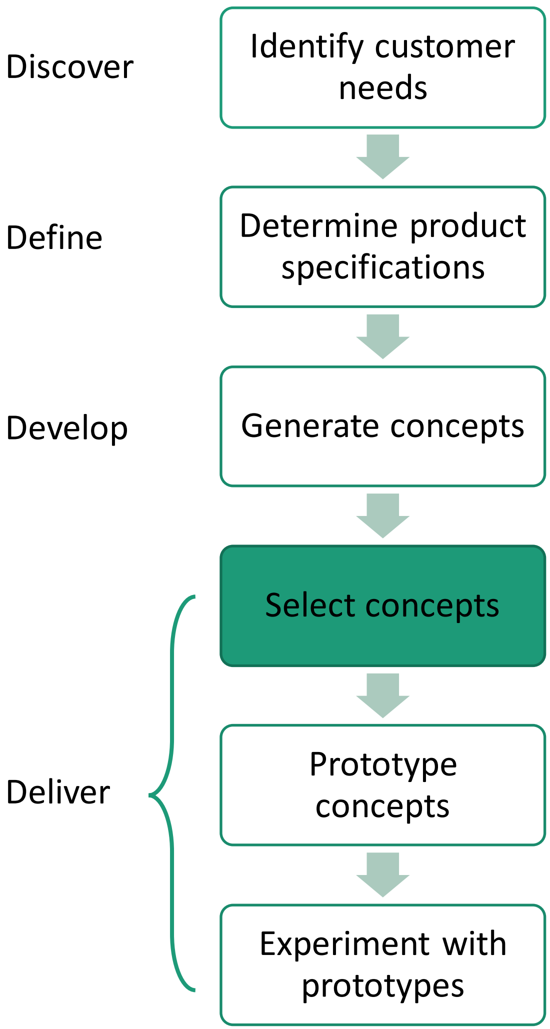
- Clearly define the problem in order to gain solutions and meet the customer's needs
- Convert customer needs to technical needs
- Establish design specifications including technical / non-technical requirements
  - Performance, maintenance, reliability
  - Cost, aesthetics, …
- Evaluate the relative importance of design parameters in relation with the product feature

Systematic Design Procedure



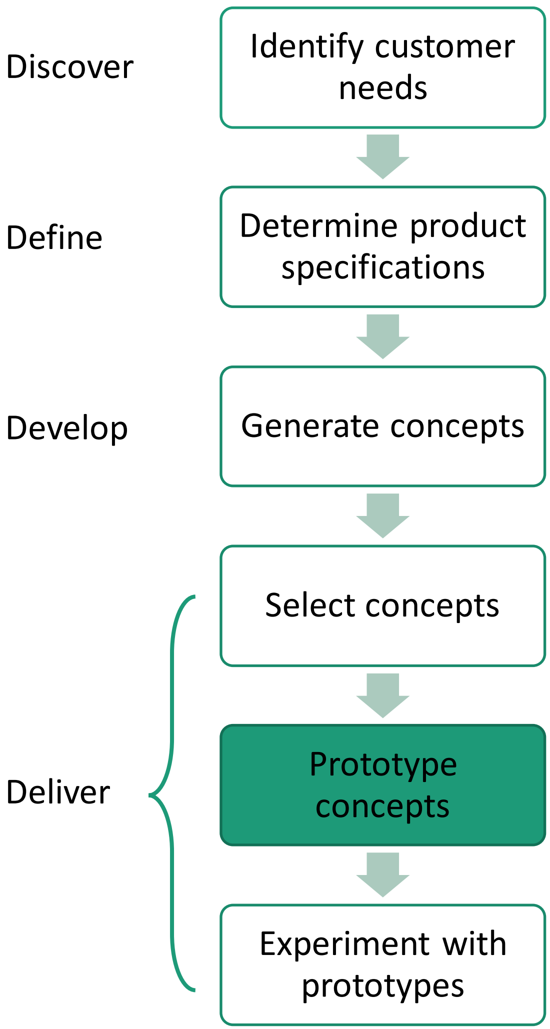
- Explore conceptual solutions that meet design specifications
- Conceptual Design = Free-hand sketch showing main feature and principle of the product
- Requires “creativity”
- Brainstorming, Biomimetric approach, TRIZ, Literature review, patent, …

Systematic Design Procedure



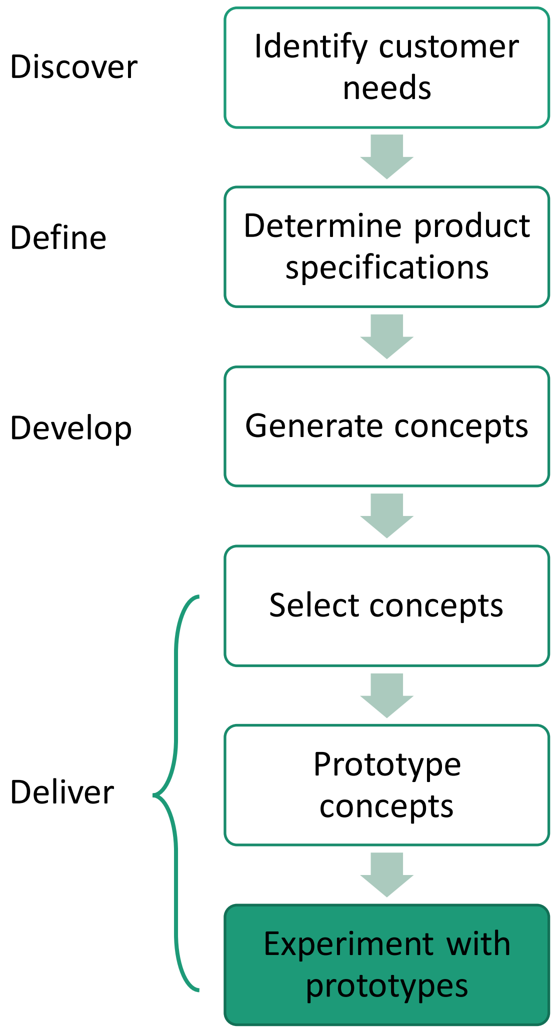
- Rough (order-of-magnitude) estimation of performance and cost of conceptual designs using mathematical model
- Evaluate conceptual designs regarding design specifications
- Select the best prospect for the conceptual design
- Bad conceptual design results in a poor final product even with good product design and manufacturing capability.

Systematic Design Procedure



- Analysis & Synthesis
- Convert conceptual design into engineering hardware by designing individual parts and assembly
- Develop specifications for main and sub parts (material, shape, surface treatment, manufacturing process, etc)
- Consider cost for each part
- Make engineering detailed drawings of all parts, and main/sub assembly for production and inspection
- CAD/CAM

Systematic Design Procedure

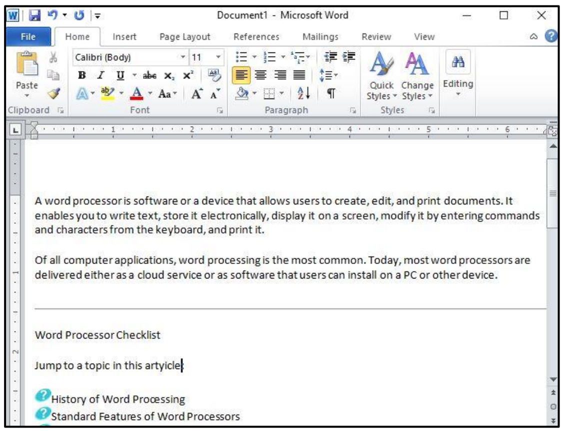


- Collect user feedback for your prototype
- Look at customer feedback to revise design and product
- Perform laboratory experiments on your prototype’s performance
- Use obtained experimental data to fine-tune the performance of your prototype
- Compile final specifications of product

Researching Users

What problems do following products solve?

Word Processor



GPS Navigation

Angry Birds

Answering Key Questions

- Who are my users?
- What are my users trying to accomplish?
- How do my users think about what they're trying to accomplish?
- What kind of experiences do my users find appealing and rewarding?
- How should my product behave?
- What form should my product take?
- How will users interact with my product?
- How can my product's functions be most effectively organized?
- How will my product introduce itself to first-time users?
- How can my product put an understandable, appealing, and controllable face on technology?
- How can my product deal with problems that users encounter?
- How will my product help infrequent and inexperienced users understand how to accomplish their goals?
- How can my product provide sufficient depth and power for expert users?

Problem Definition

- Formal Description of the task (mission) to be solved
- Also referred to as ‘problem statement’, ‘mission statement’
- Formal
  - In written form
  - Approved by appropriate people (client, manager, etc.)
- Starting point of overall engineering design process

Problem Definition

- Problem = task to be solved
- Do not describe difficulties or weakness as problem definition

  ➔ Describe what to do for difficulties or weakness

  eg) Current railway transportation is very slow (X)

  Improve the speed of railway transportation (O)
- Do not use subjective and relative terms in problem definition

  ➔ Describe objective and verifiable terms

  eg) Build an archiving system with large storage (X)

  Build an archiving system with storage of at least 5TB (O)

Problem Definition

- Purpose of problem definition
  - To clarify the purpose of design among many stakeholders (clients, users, managers, developers, etc.)
  - To provide guidelines for engineering design work
  - To verify the success of final design products
- Problem definition can be hierarchical
  - Problem Definition for the whole project or whole teams
  - Problem Definition for sub-projects or particular team

Early Validation is Important

- Validate your problem
  - Ensure that the problem exists in real life, not in your fantasy world no matter how fabulous and innovative is your product idea.
- Validate your market
  - Ensure that the problem is bad enough, so people are willing to pay for its solution
  - Start with smaller markets with more specific problems. You will expand later.
- Validate your product
  - Ensure that your product can really solve identified problem for the specified market.

Steps for Early Validation

1. Define your target market in specific terms. For example:
   - Urban moms who want to save on their grocery purchases
   - People who spend more than 4 hours a day on Instagram and play at least 3 different social games
   - Avoid validating your problem, market and product with your friends, family or random strangers
2. Find 5 people who belong to your target market and visit them at their homes, offices or whereever your expect them to use your product. Answer following questions after your visit:
   - Is there a specific problem encountered by each participant?
   - Have they all complained about a particular piece of software?
   - Are they performing tasks in a specific way that could be made easier?

   Never ever try to sell your participants your product idea.
   - It will bias them.
   - You are there to listen and observe their surroundings.
   - You are there to validate that your problem exists.

Steps for Early Validation

3. Make a landing page which advertises your yet not existing product.
    - Make a free website in LaunchRock or a simple Facebook page OR hire decent designers from places like 99Designs, Fiver, or Elance.
    - Promote your landing page with AdWords, Facebook ads or using other SMM techniques.
    - Hook up some Google Analytics to count clicks on your “Buy”, “Subscribe”, “Pre-order” buttons.
    - Check how many people visit, subscribe or even make reservations for your fictional product.
    - You are doing this to validate your market.
4. Make prototypes of your product and show them to your potential customers
    - Explaining the concept of your product and asking whether it solves their problem is the **worst thing to do** to your potential customers.
    - Instead, show them something OR give them something to play with, and **observe their reactions.**
    - You are doing this to validate your product.

Researching Users

- User research covers a wide range of topics starting from doing ethnographic interviews with your target group, to classical usability studies, to quantitative measurements on your user experience design.
- Qualitative research methods are exploratory and seek to get an in-depth understanding of the experiences and everyday lives of individual users.
- Quantitative research methods seek to measure user behaviour in a way that can be quantified and used for statistical analysis.

Qualitative Research: Ethnographic Interviews

- Ethnography is a term borrowed from anthropology, which means systematic and immersive study of human cultures.
- Propose a behavioural model and find real candidates to study
- Organize interviews with two designers: moderator and facilitator.
- Key guidelines:
  - Interview the interaction happens
  - Avoid fixed set of questions
  - Assume the role of an apprentice, not an expert
  - Use open-ended and closed-ended questions to direct the discussion.
  - Focus on goals first, and tasks second
  - Avoid making the user a designer
  - Encourage storytelling
  - Ask for a shown-and-tell
  - Avoiding leading questions

Qualitative Research: Competitor Tests

- Find some users of your competitors (don’t say that your product is unique and you have no competitors)
- Observe how they use your competitor’s product.
- Ask questions like:
  - What are your likes/dislikes about this product?
  - How did you learn to use it?
  - What confuses you in this product?
  - Have you tried anything else to mitigate this difficulty?

Qualitative Research: Five-Second Tests

Show your potential users your landing page and let them scroll through it. Here is how:

1. Go to your local coffee shop, restaurant or other public place without looking too creepy or getting arrested.
2. Bring your tablet/laptop with few version of your landing page… and some cash.
3. Ask people if they will look at a couple of screens for your startup in exchange for a cup of coffee/chocolate bar/etc.
4. Then ask following questions:
   - What does this product do?
   - Who is this product for?
   - What would you do if you cam to this page on the recommendation of a friend?
5. Feel free to follow up with polite question what makes them think those things.

OR you can use UsabilityHub for the same purpose.

Qualitative Research: Clickable Prototype Testing

Show your potential users some interactive UI without any backend logic. User can switch between your mock screens as she clicks.

1. Make an interactive prototype using Figma, Axure or even simple HTML, CSS and JavaScript.
2. Create list of tasks for your users to perform on your prototype. For example:
   - Sign-up flows that have more than one step
   - Purchase flows
   - Searching and browsing experiences, e.g. find a particular product.
   - Sharing experiences, e.g. take a picture/leave a comment
   - File uploads and editing
   - Navigation of the entire product
   - Anything else that requires more than 2 steps
3. Ask 3 to 5 potential users to perform these tasks on your prototype
4. Identify problems that users had while completing the tasks, fix them in your prototype and do it all again.
5. Keep iterating until users are able to complete most tasks without crying.

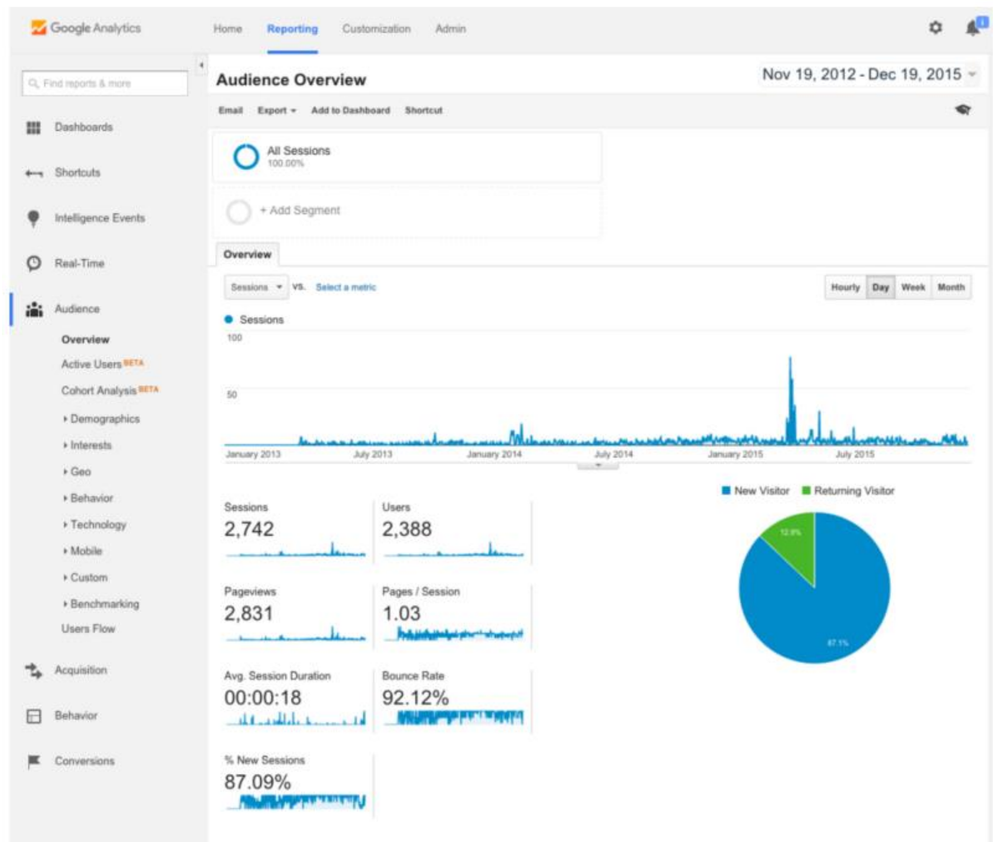
Quantitative Research

- Insight-driven research seeks to understand: what the problem space is, why the problem exists; where opportunities lie.
- Evaluative research measures how a particular design stands up to its KPIs. It helps to answer how the user flow is effective both before and after proposed changes.
- Generative research offers opportunities to create and explore new designs through research.

Quantitative Research: System Analytics

Google Analytics

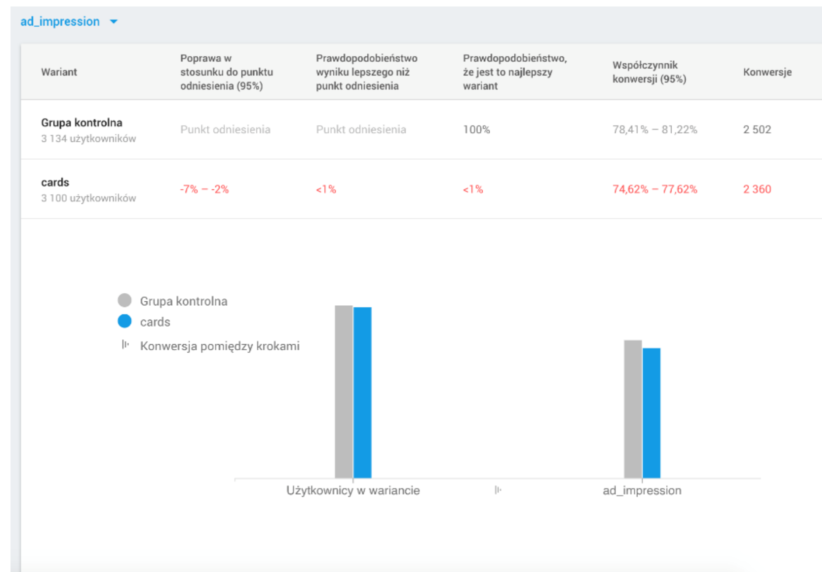
- Easy and free to setup on your landing page.
- Provides insights on users flow, demographics, and geography
- Helps to understand how customers access and navigate your website.
- Learn More: [Google Analytics]



Quantitative Research: A/B Testing

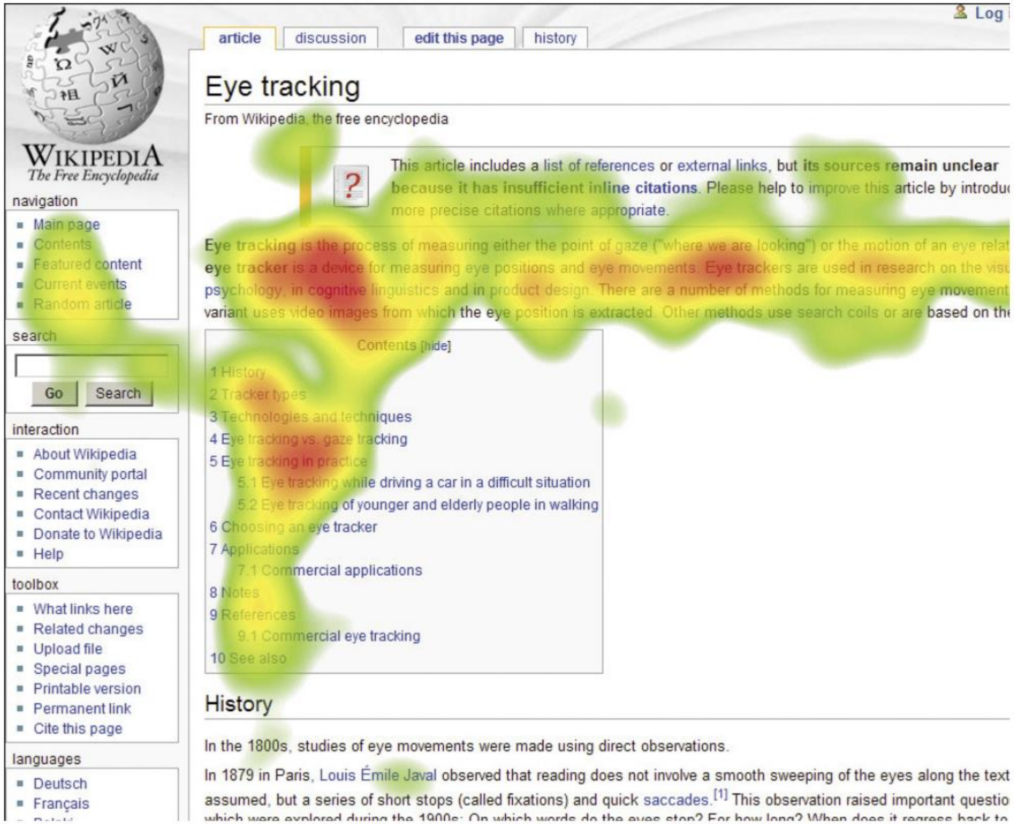
- It displays one of two alternative solutions to randomly selected customer groups, and determines preferred one.
- For that, website/mobile app should be hosted on a special platform and analytics engine should be hooked up.
- Various experiments can be created and their results can be investigated.
- Learn more: [Firebase A/B Testing]

Firebase A/B Testing

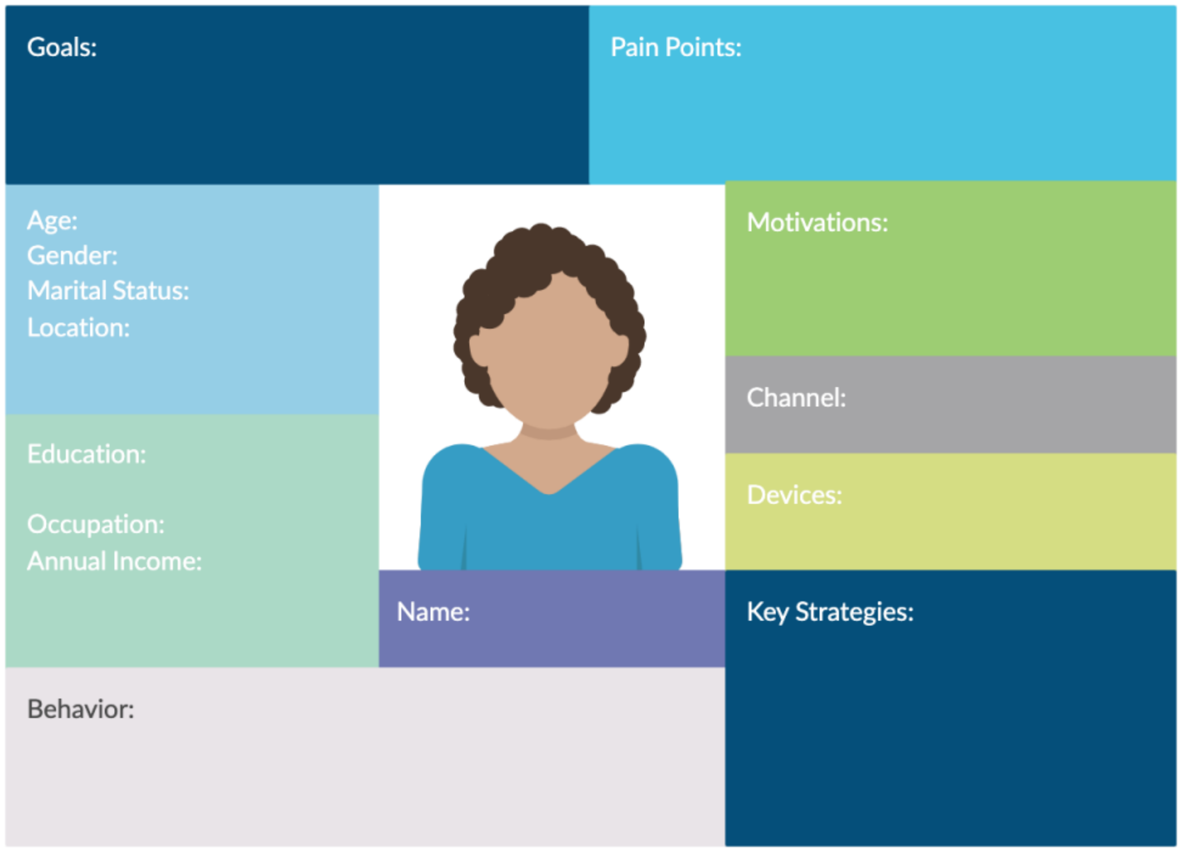


Quantitative Research: Eye Tracking

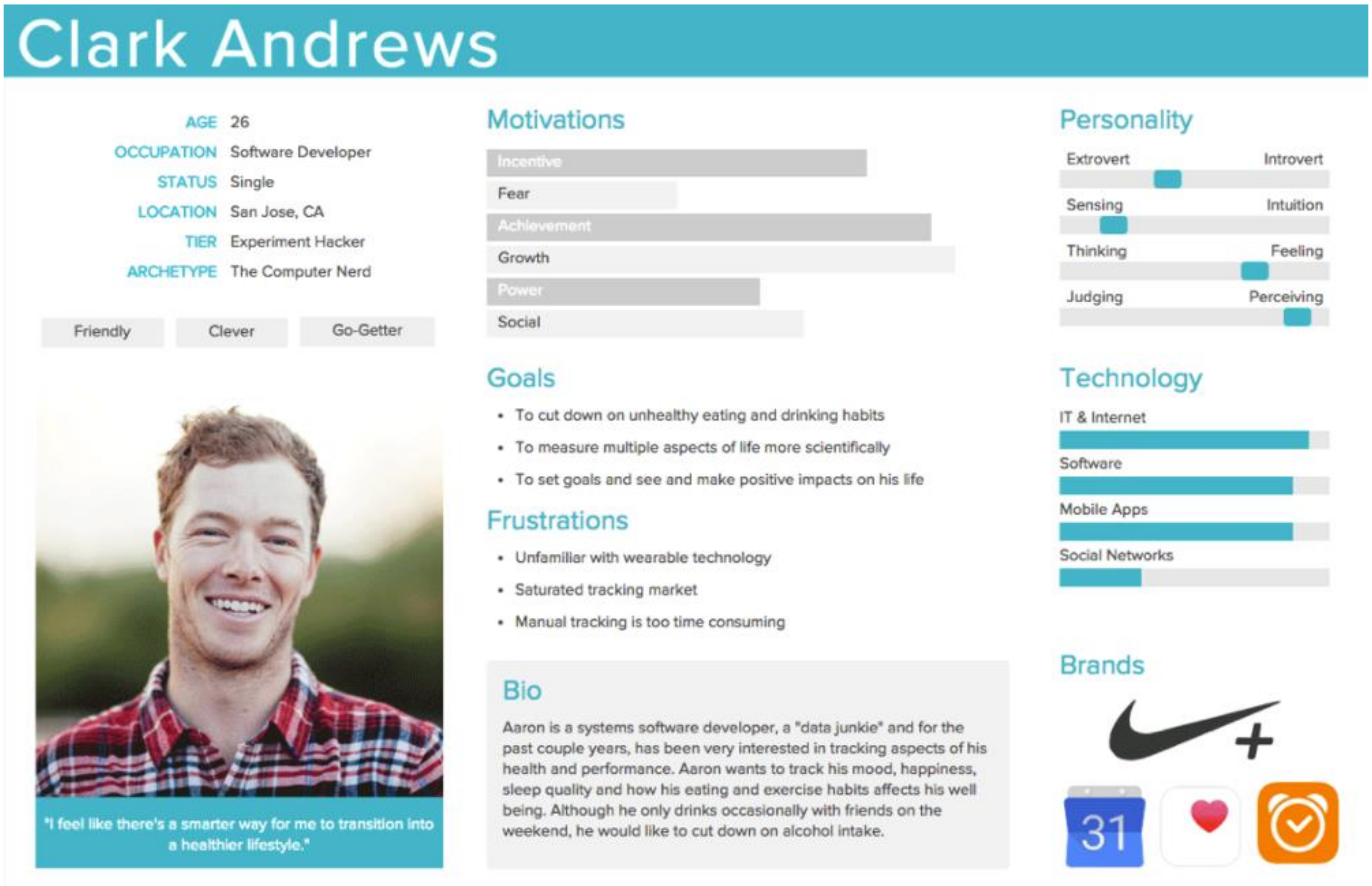
- It is the process of using cameras to follow a participant's eyes as they scan a page.
- It may be a major hurdle, because participants must be brought into a lab.
- It may be expensive.
- Tobii is a leader in building eye-tracking software.



Persona Profiles

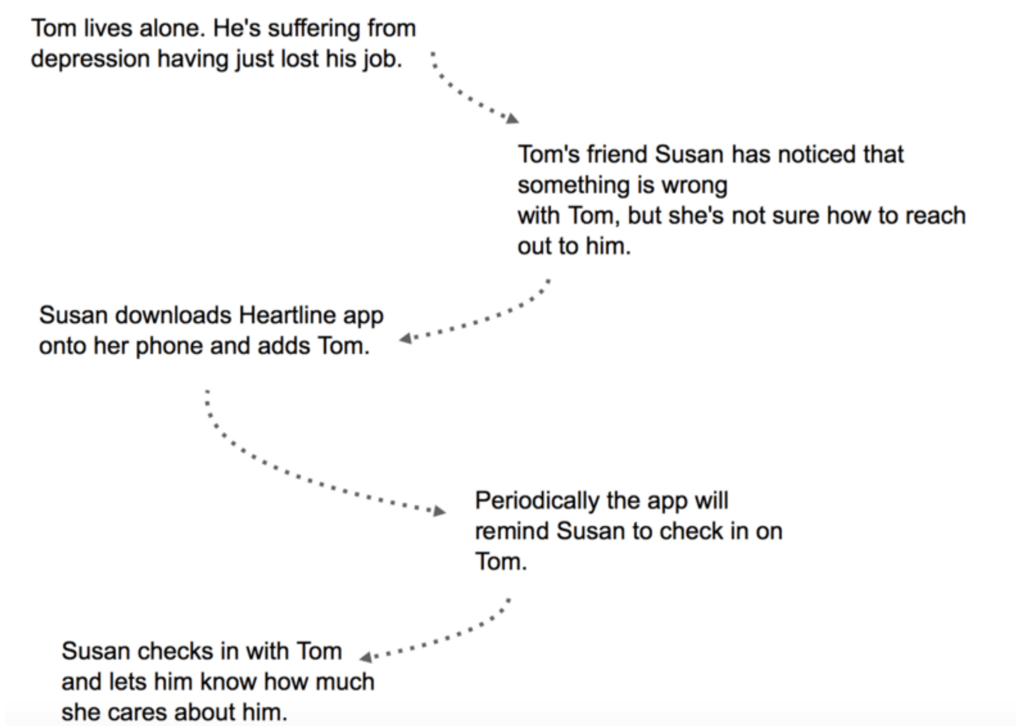


Persona Example



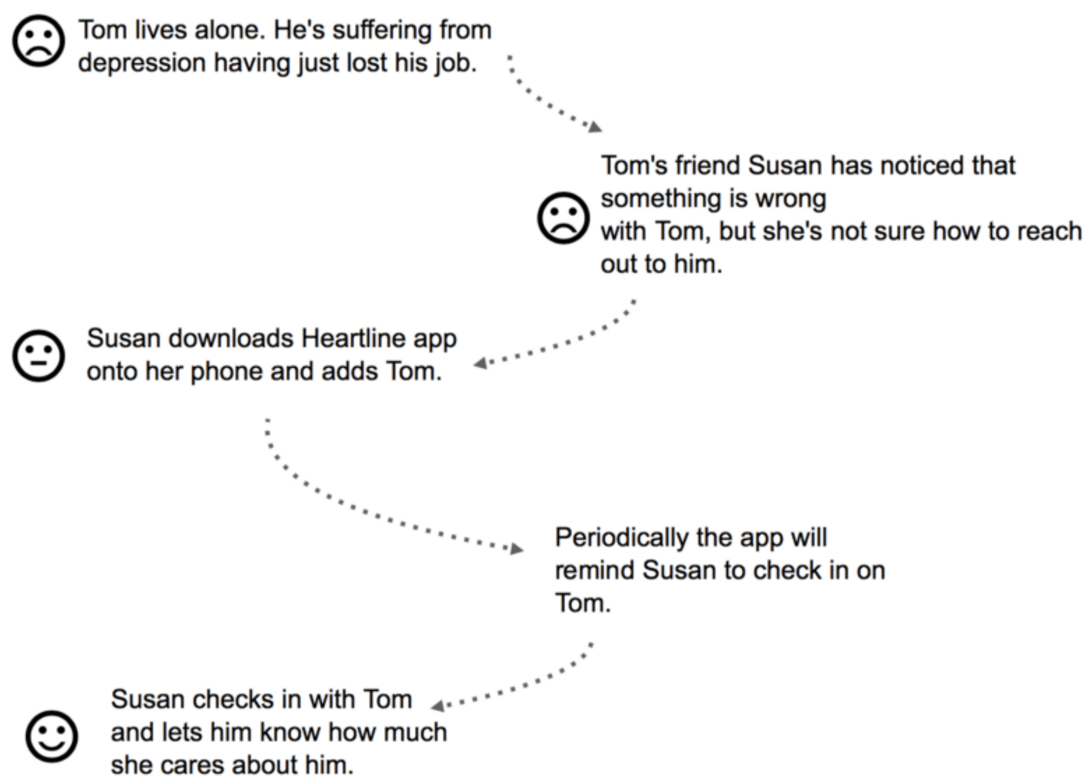
Storyboarding

1. Grab a pen and paper.
2. Start with a plain text and arrows.



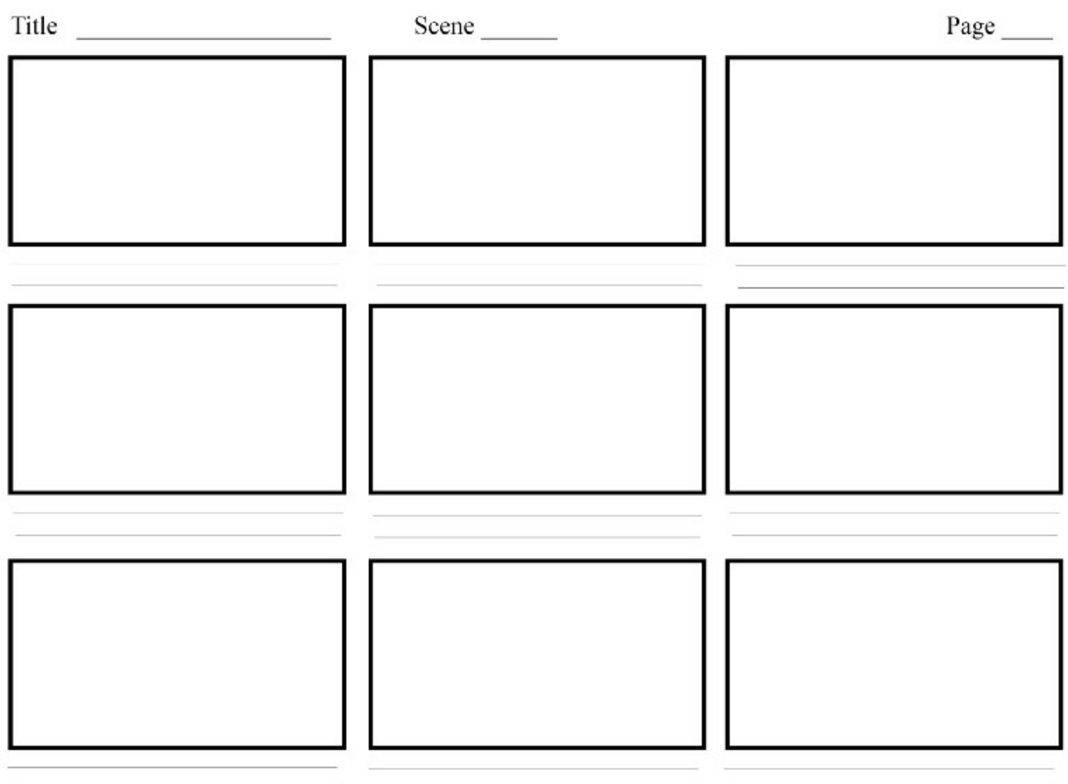
Storyboarding

3. Bake emotion into the story

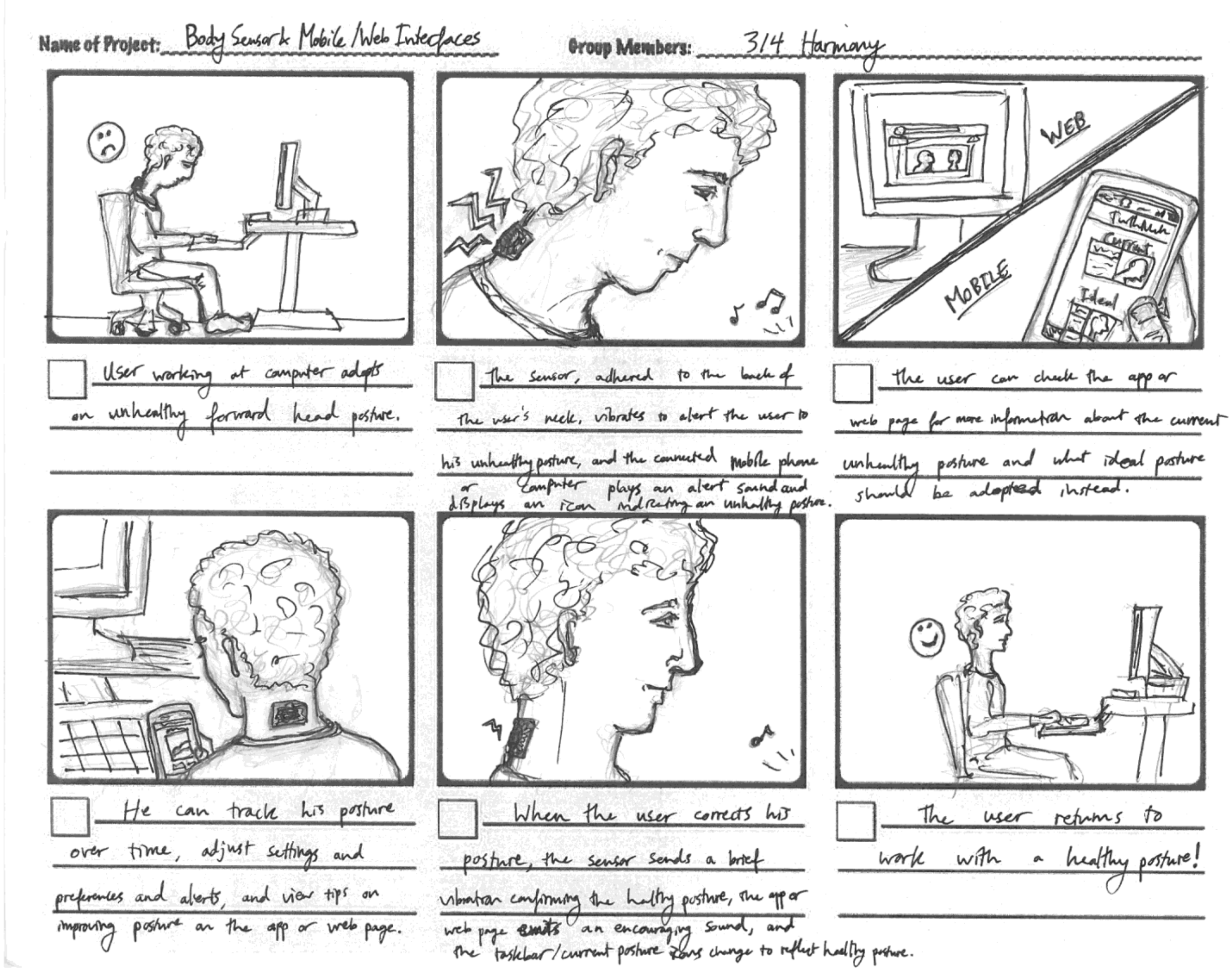


Storyboarding

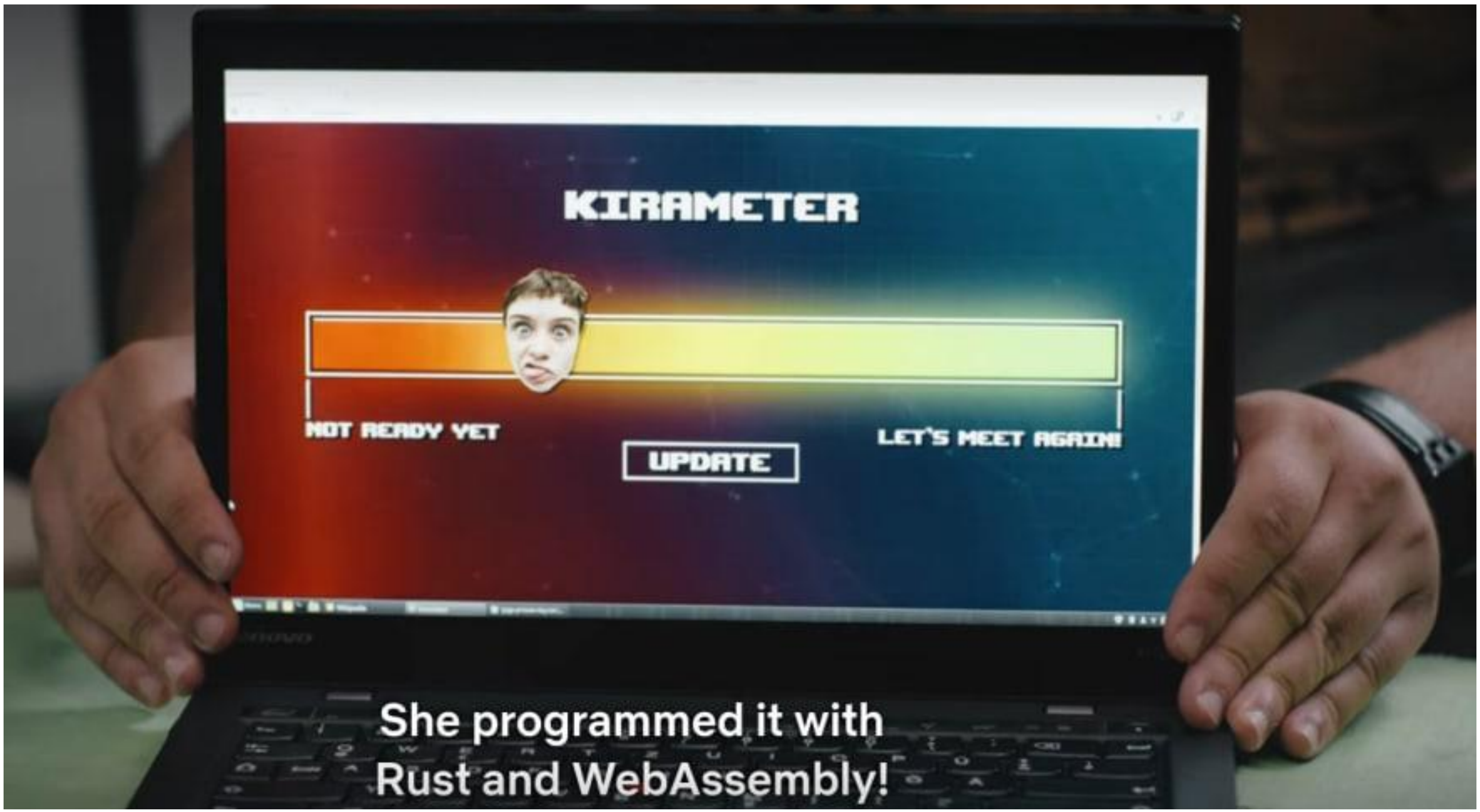
4. Translate each step into a frame.



Storyboarding



One Possible Concept of Heartline



Screen from movie “How to Sell Drugs Online (fast)”

Storyboarding Exercise

- You need to create a concept of a product that can help people with bad eye-sight.
- Below are the steps:
  1. Propose your initial concept of a product for people with bad eye-sight
  2. Create 1 to 3 personas and tell a story involving them (using MidJourney)
  3. Write your story in short paragraphs (using ChatGPT)
  4. Ensure that your story has an emotional drama
  5. Mark each paragraph with corresponding smiley
     - btw, you can use your own custom smileys for unique life situations
  6. Identify the culmination of your story. Think how your product may help in this situation and make a dramatic turn in persona’s life.
  7. Adapt and place your product into the storyline
  8. Draw sketches of each scene in your story (using MidJourney)

Identifying Customer Needs

Identifying Customer Needs

- Ensure that the product is focused on customer needs
- Identify latent or hidden needs as well as explicit needs
- Provide a fact base for justifying the product specifications
- Create an archival record of the needs activity of the development process
- Ensure that no critical customer need is missed or forgotten
- Develop a common understanding of customer needs among members of the development team

Steps of Identifying Customer Needs

1. Define the Scope
   - Mission Statement
2. Gather Raw Data
   - Interview
   - Focus Groups
   - Observation
3. Interpret Raw Data
   - Need Statements
4. Organize the Needs
   - Hierarchy (primary, secondary and tertiary needs)
5. Establish Importance
   - Surveys
   - Quantified needs
6. Reflect on the Process
   - Continuous improvement

Customer Needs Example: Smart Thermostat

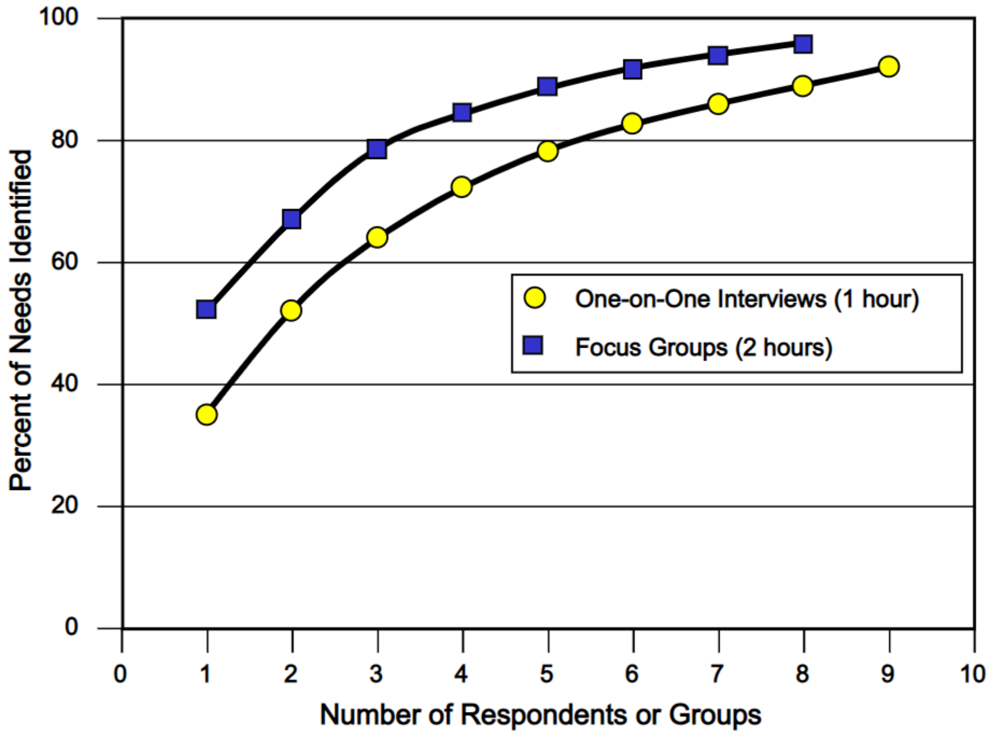
1. Define the Scope

Mission Statement: Thermostat Project	
Product Description	• Thermostat for residential use.
Benefit Proposition	• Simple to use, attractive, and saves energy.
Key Business Goals	• Product introduced in fourth quarter of 2012. • 50% gross margin. • 10% share of replacement thermostat market by 2016.
Primary Market	• Residential consumer.
Secondary Markets	• Heating, ventilation, and air-conditioning contractors doing residential work.
Assumptions	• Replacement for an existing thermostat. • Compatible with most existing systems and wiring.
Stakeholders	• User • Retailer • Sales force • Service center • Production • Legal department

2. Gather Raw Data

- From the use environment of the product
  - Interviews
  - Focus Groups
  - Observing the product in use
  - Online Surveys?
- Choosing customers
  - Quality over quantity
  - Concept development teams
  - Lead users
    - Articulate emerging needs because of inadequacies
    - May have invented solutions
  - Extreme users
    - Users with extreme/non-conventional needs

How many customers?



From: Griffin, Abbie and John R. Hauser. "The Voice of the Customer", *Marketing Science* . vol. 12, no. 1, Winter 1993.

- Usually interviewing is cheaper and takes place in use environment, hence it is more recommended.
- Gathering needs should continue until there are no more new needs identified.
- Usually 10 to 30 hours of interviewing and focus group sessions will cover 90% of customer needs.

Eliciting customer needs

- When and why do you use this product?
- Walk us through a typical session using the product?
- What do you like about the existing products?
- What do you dislike about the existing products?
- What issues do you consider when purchasing the product?
- What improvements would you make to the product?

Hints for Effective Interaction with Customers

- Go with the flow
  - Don’t stick to your interview questions
- Use visual stimuli and props
  - Bring all your product collection, competitor’s products, or products that tangentially related to your product
- Suppress preconceived hypotheses about the product technology.
- Have the customer demonstrate the product and/or typical task related to the product
- Be alert for surprises and the expression of latent needs.
- Watch for nonverbal information.
- Data privacy

Documenting Interaction with Customers

- Audio recording
- Video recording
- Photography
- Notes

Customer:	Bill Esposito	**Interviewer(s):**	Jonathan and Lisa
Address:	100 Memorial Drive Cambridge, MA 02139	**Date:**	19 January 2018
Telephone:	617-864-1274	**Currently uses:**	Honeywell Model A45
E-mail:	bespo@zmail.com	**Type of user:**	Homeowner
Willing to do follow-up?	Yes		

Question/Prompt	Customer Statement	Interpreted Need
Typical uses	I have to manually turn it on and off when it gets too hot or cold.	The thermostat maintains a comfortable temperature without requiring user action.
	Each time I want to change the temperature, I need to adjust both thermostats in the house.	Any user inputs need not be made in multiple locations. (!)
Likes—current model	I like that I can change the temperature if the setting is too high.	The temperature setting is easy to control manually.
	It didn't cost a fortune.	The thermostat is affordable to purchase.
Dislikes—current model	I'm too lazy to learn how to figure it out.	The thermostat requires little or no user instruction or learning.
	I sometimes forget to turn it off when we leave the house.	The thermostat saves energy when no one is home. (!)
	Sometimes the buttons don't register a push and I have to press it repeatedly.	The thermostat definitively registers any user inputs.
Suggested improvements	I would like my iPhone to adjust my thermostat.	The thermostat can be controlled remotely without requiring a special device.
	I like to shift quickly between different options like energy saving or ultra comfort.	The thermostat responds immediately to differences in occupant preferences.

3. Interpret Raw Data in Terms of Customer Needs

- Express the need in terms of what the product has to do, not in terms of how it might do it.
- Express the need as specifically as the raw data
- Use positive, not negative, phrasing
- Express the need as an attribute of the product
- Avoid the words *must* and *should*

Example of expressing customer needs

Guideline	Customer Statement	Needs Statement—Right	Needs Statement—Wrong
"What" not "How"	I would like my iPhone to adjust my thermostat.	The thermostat can be controlled remotely without requiring a special device.	The thermostat is accompanied by a downloadable iPhone app.
Specificity	I have different heating and cooling systems.	The thermostat can control separate heating and cooling systems.	The thermostat is versatile.
Positive not negative	I get tired of standing in front of my thermostat to program it.	The thermostat can be programmed from a comfortable position.	The thermostat does not require me to stand in front of it for programming.
An attribute of the product	I have to manually override the program if I'm home when I shouldn't be.	The thermostat automatically responds to an occupant's presence.	An occupant's presence triggers the thermostat to automatically change modes.
Avoid "must" and "should"	I'm worried about how secure my thermostat would be if it were accessible online.	The thermostat controls are secure from unauthorized access.	The thermostat must be secure from unauthorized access.

Needs Translation Exercise

- I don’t want to lose a fortune on gas bills every month
- Every time before going to bed, I set the temperature to 20 C, and wake up in the middle of the night from nauseating heat.
- I don’t want to heat my room when it is hot outside
- Seasons are unpredictable in my country, early winters can be hot and late summers can be cold. It would be nice to have a thermostat which could adapt to seasons of the year automatically.
- My kids like to play with thermostat and occasionally they heat up the room without our notice.
- I am too old to understand these new touchscreen devices. I find using them quite hard to people of my age.
- I want to set the temperature while my kids are sleeping at night and not wake them up in the process.

4. Organize List of Customer Needs

- You may gather 50 to 300 needs
- Print them in cards/sticky notes
- Eliminate redundant statements
- Group cards according to their similarity from customer's perspective
- Each group can be represented with some customer need as its label
- Grade needs with "*"s depending on their importance to the customer. Label latent needs with "!".

**	**The thermostat is easy to install.**
***	The thermostat works with my existing heating and/or cooling system.
**	The thermostat installation is an easy do-it-yourself project for a novice.
**	The thermostat can control separate heating and cooling systems.
*	The thermostat can be installed without special tools.
	The thermostat is easily purchased.
*	**The thermostat lasts a long time.**
	The thermostat is safe to bump into.
	The thermostat resists dirt and dust.
!	The thermostat exterior surfaces do not fade or discolor over time.
	The thermostat is recyclable at end of life.
***	**The thermostat is easy to use.**
**	The thermostat user interaction is easy to understand.
*	The thermostat is easy to learn to use.
*	The thermostat does not place significant demands on user memory.
!	The thermostat can be programmed from a comfortable position.
	The thermostat can be controlled remotely without requiring a special device.
!	The thermostat works pretty well right out of the box with no setup.
	The thermostat's behavior is easy to change.
	The thermostat is easy to control manually.
	The thermostat display is easy to read from a distance.
	The thermostat display can be read clearly in all conditions.
	The thermostat's controls accommodate users with limited dexterity.
	The thermostat accommodates different conventions for temperature scales.
	The thermostat accommodates different preferences for representing time and date.
**	**The thermostat controls are precise.**
**	The thermostat maintains temperature accurately.
	The thermostat minimizes unintended variability in temperature.
	The thermostat allows temperatures to be specified precisely.
***	**The thermostat is smart.**
***	The thermostat can adjust temperature during the day according to user preferences.
**	The thermostat can be programmed to a precise schedule.
!	The thermostat automatically responds to occupancy.
!	The thermostat prevents pipes from freezing in cold months.
	The thermostat alerts the user when a problem arises.
	The thermostat does not require users to set time or date.
	The thermostat adjusts automatically to the seasons.
*	**The thermostat is personal.**
*	The thermostat accommodates different user preferences for comfort.
	The thermostat accommodates different user preferences for energy efficiency.
	The thermostat controls are secure from unauthorized access.
	The thermostat provides useful information.
***	**The thermostat is a good investment.**
**	The thermostat is affordable to purchase.
***	The thermostat saves energy.
*	The thermostat tracks cost savings.
**	**The thermostat is reliable.**
	The thermostat does not require replacing batteries.
	The thermostat works normally when electric power is suspended.

5. Establish relative importance

Thermostat Survey

For each of the following thermostat features, please indicate on a scale of 1 to 5 how important the feature is to you. Please use the following scale:

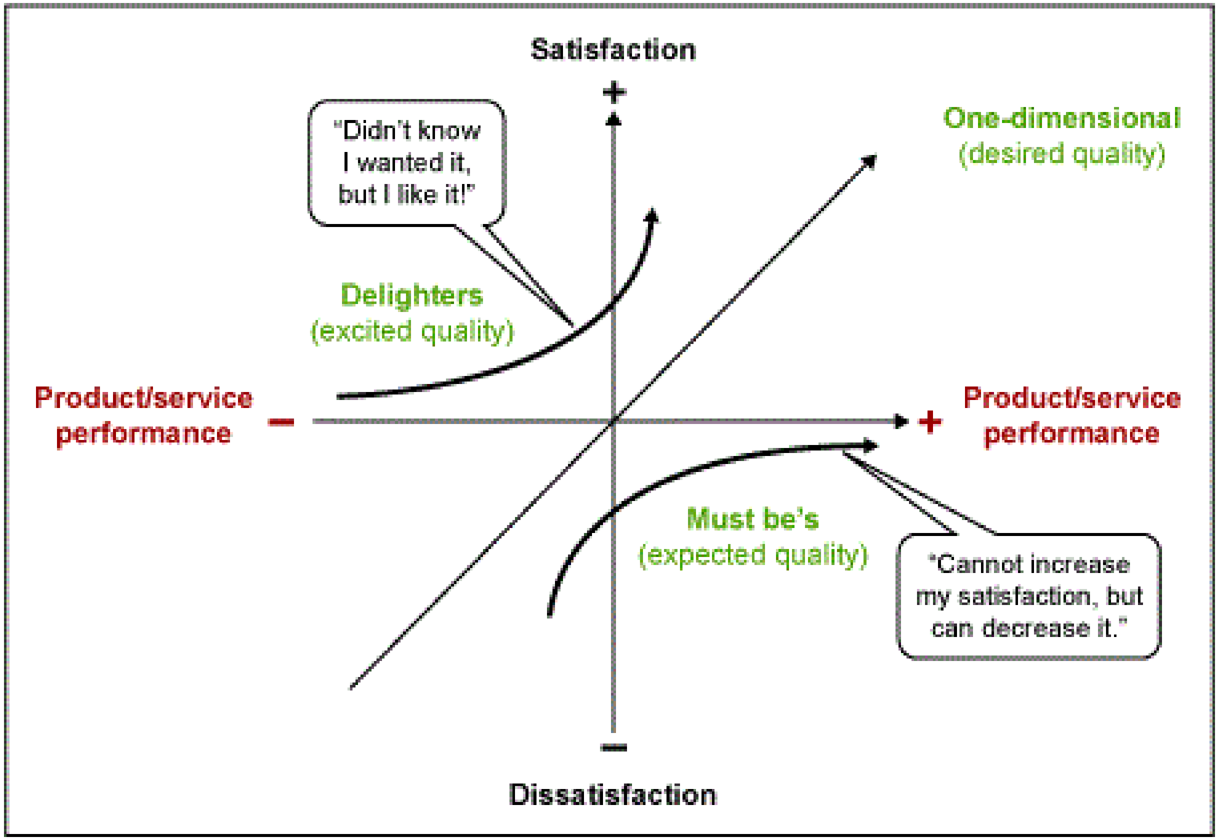
1. Feature is undesirable. I would not consider a product with this feature.
2. Feature is not important, but I would not mind having it.
3. Feature would be nice to have, but is not necessary.
4. Feature is highly desirable, but I would consider a product without it.
5. Feature is critical. I would not consider a product without this feature.

Also indicate by checking the box to the right if you feel that the feature is unique, exciting, and/or unexpected.

Importance of feature on scale of 1 to 5		Check box if feature is unique, exciting, and/or unexpected.
_______	The thermostat does not require the user to set time or date.	☐
_______	The thermostat does not require replacing batteries.	☐
_______	The thermostat adjusts automatically to the seasons.	☐
_______	The thermostat accommodates different preferences for representing date and time.	☐

And so forth.

Kano Diagram



6. Reflect on the Results and the Process

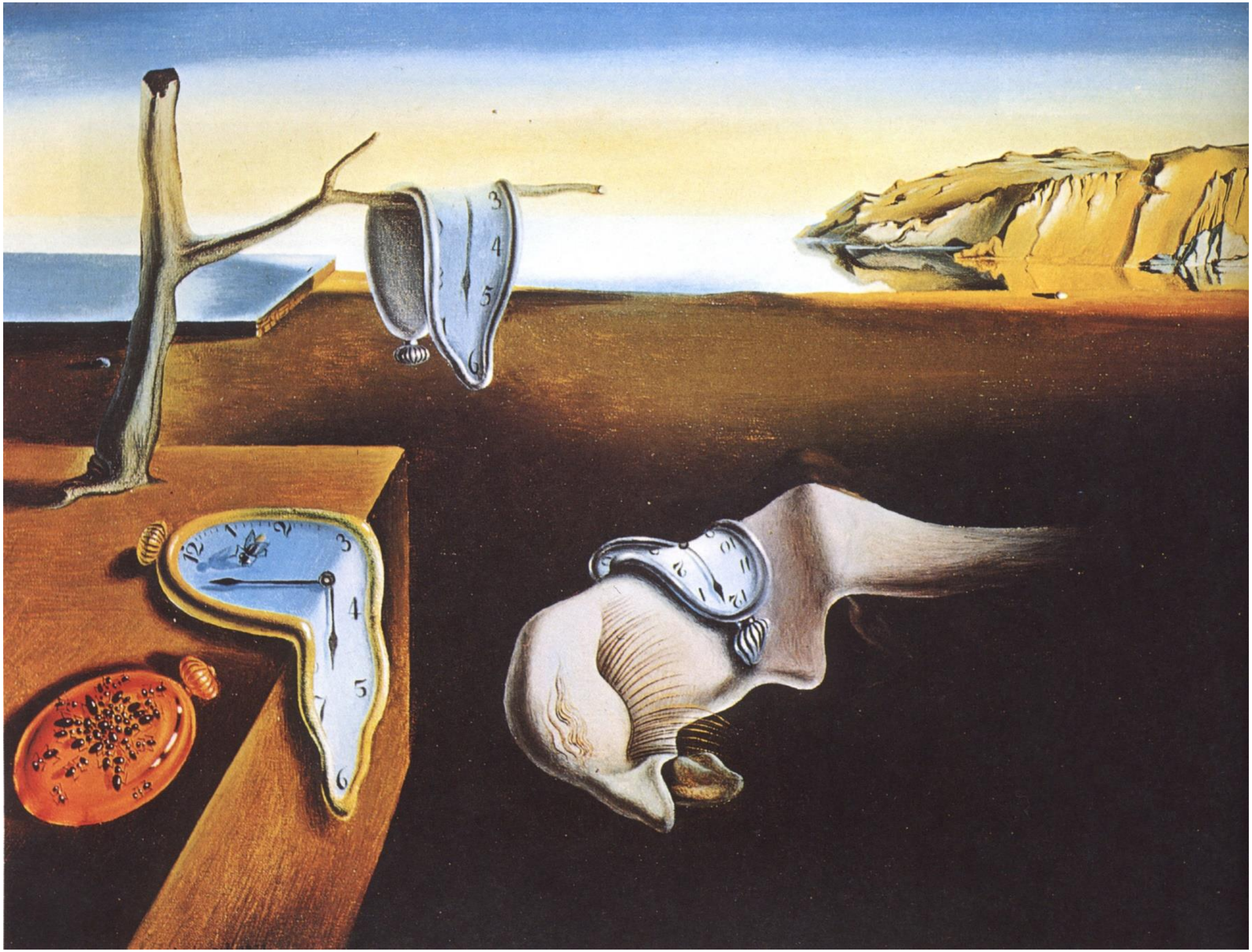
- Have we interacted with all of the important types of customers in our target market?
- Are we able to see beyond needs related only to existing products to capture the latent needs of our target customers
- Which of the customers we spoke to would be good participants in our ongoing development efforts?
- What do we know now that we didn’t know when we started? Are we surprised by any of the needs?
- How might we improve this process of eliciting needs in future?

Summarize Your Results

- Basic Needs:
  - Thermostat is easy to use
  - Thermostat can adjust temperature during the day according to user preferences
  - Thermostat works with my existing heating/cooling system
  - Thermostat reduces energy consumption
- Latent Needs:
  - Thermostat can be programmed from a comfortable position
  - Thermostat works pretty well right out of the box with no setup
  - Thermostat automatically responds to occupancy
  - Thermostat exterior surfaces do not fade or discolor over time
  - Thermostat prevents pipes from freezing in cold months

Ant Colony Optimization

Engineering Idea #2

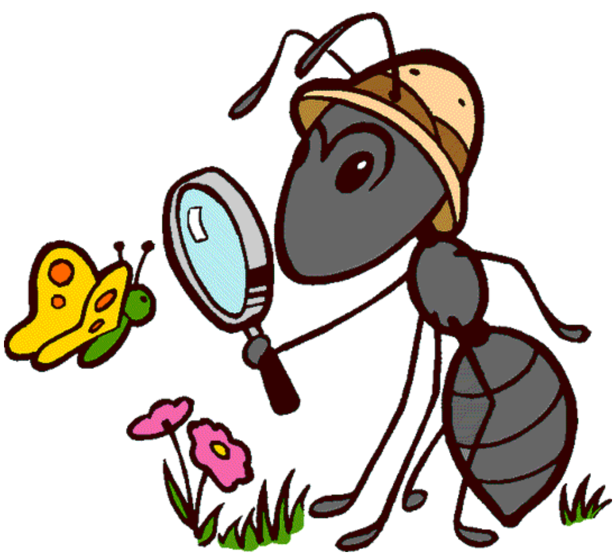


Painting by Salvador Dali (1934): “The Persistence of Memory”, Museum of Modern Art, NY

Outline

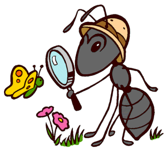
1. **Biological inspiration of ACO**
2. Solving NP-hard combinatorial problems
3. ACO for the Traveling Salesman Problem

Biological inspiration:
from real to artificial ants



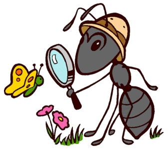
Ant colonies

- Distributed systems of social insects
- Consist of **simple** individuals
- Colony intelligence >> Individual intelligence



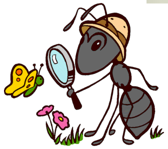
Ant Cooperation

- **Stigmergy** – indirect communication between individuals (ants)
- Driven by environment modifications
- **Some example of stigmergy in ants:**
  - efficiently distribute labor: *builder*, *forager*, *soldier* or *nurse;*
  - regulation of nest temperature within 1 degree Celsius range;
  - forming bridges;
  - raiding specific areas for food;
  - building and protecting nest;
  - sorting brood and food items;
  - cooperating in carrying large items;
  - emigration of a colony;
  - **finding shortest route from nest to food source**;
  - preferentially exploiting the richest food source available.

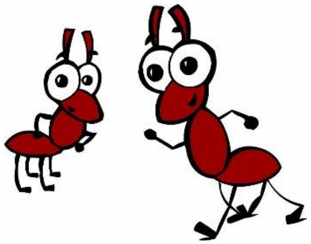


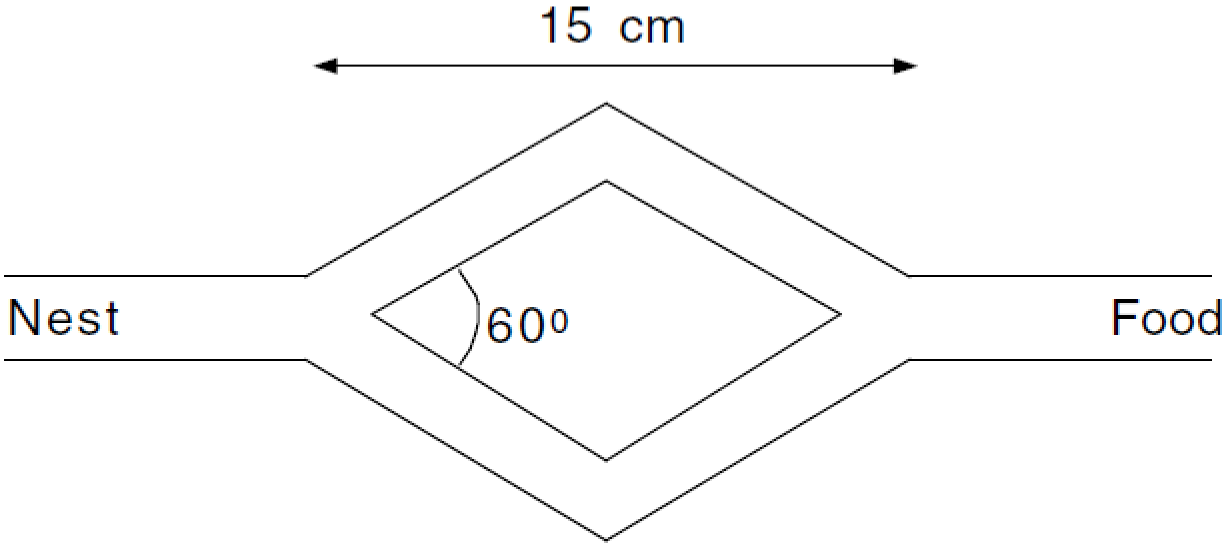
Denebourg’s double bridge experiments

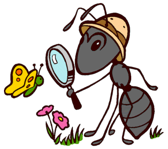
- Studied Argentine ants *linepithema humilis*
- Double bridge from ants to food source



Double bridge experiments: equal lengths (1)

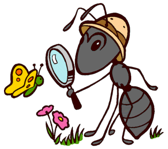




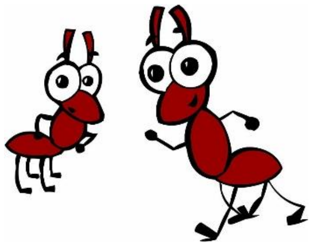


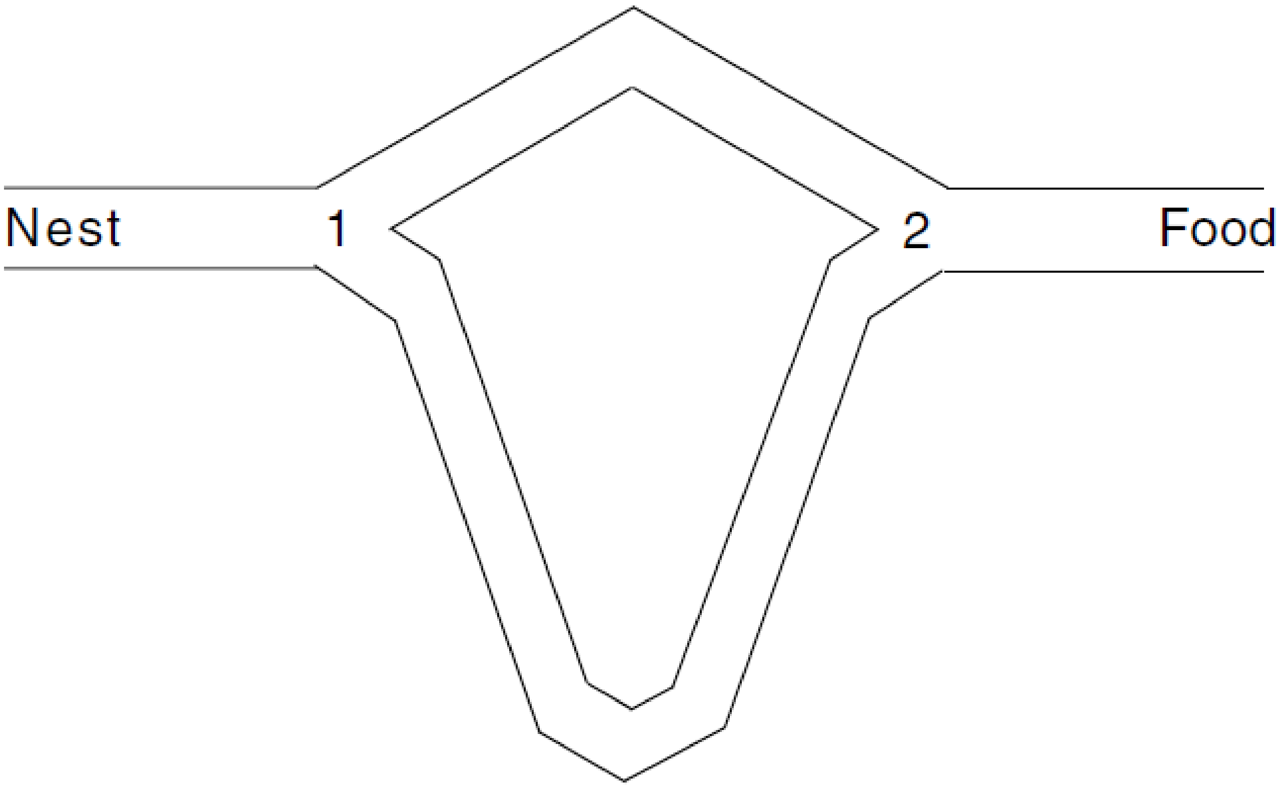
Double bridge experiments: equal lengths (2)

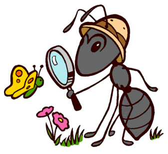
- Run for a number of trials
- Ants choose each branch ~ same number of trials



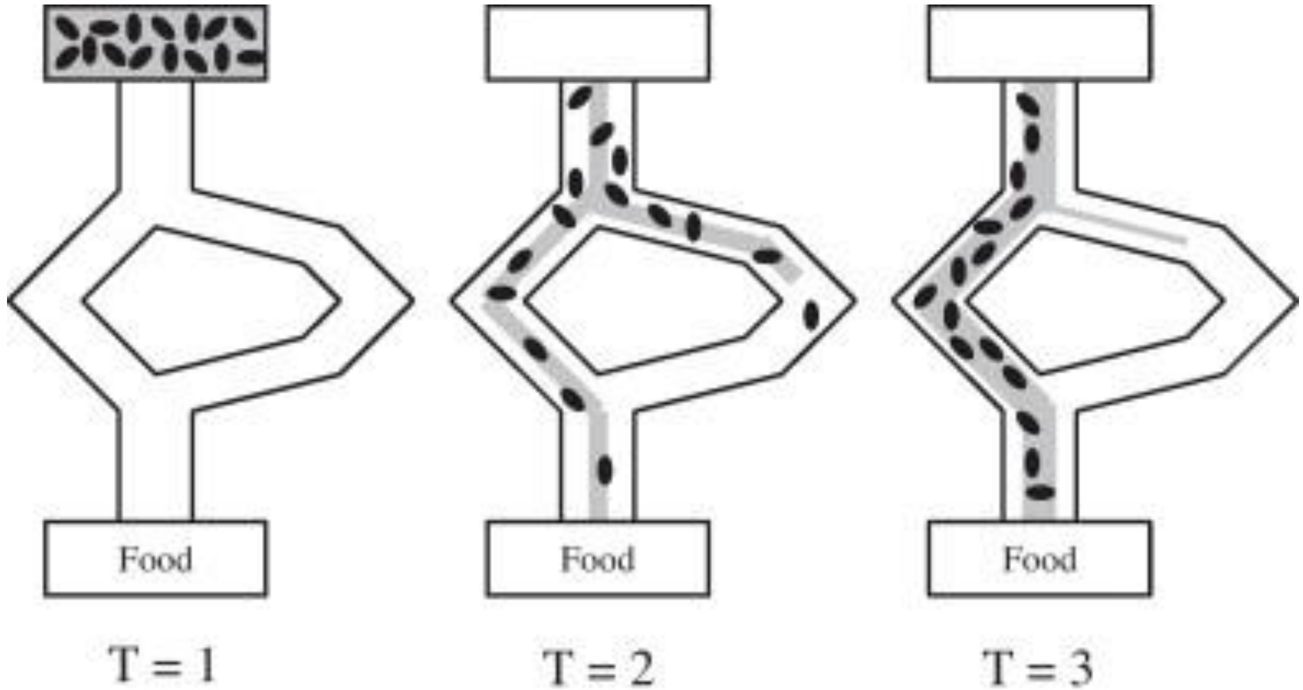
Double bridge experiments: different lengths (2)



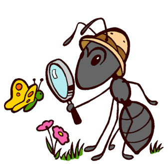




Double bridge experiments: different lengths (2)



- The majority of ants follow the short path



Outline

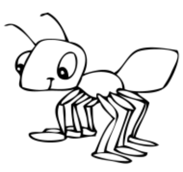
1. Biological inspiration of ACO
2. **Solving NP-hard combinatorial problems**
3. ACO for the Traveling Salesman Problem

Combinatorial optimization

- Find values of discrete variables
- Optimizing a given **objective function**

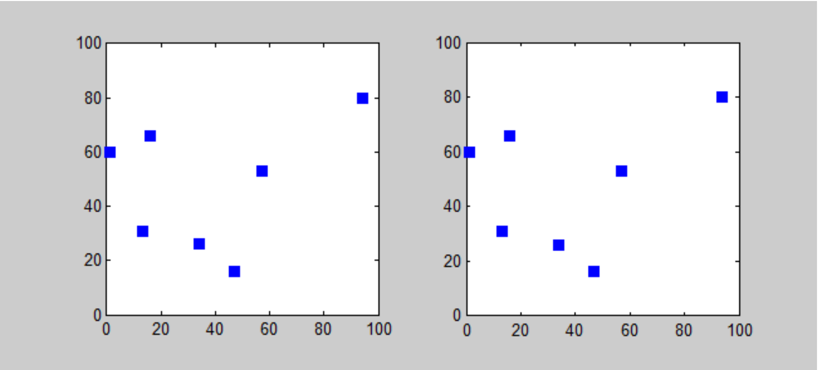
$\Pi = (S, f, \Omega)$ – problem instance

- S – set of candidate solutions
- f – objective function
- Ω – set of constraints
- $\tilde{S} \subseteq S$ – set of feasible solutions (with respect to Ω)
- Find globally optimal feasible solution $s*$



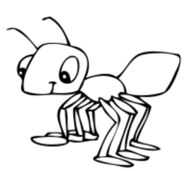
Example: Travelling Salesman Problem

- S – sequence of cities visited
- f – cost of a tour
- Ω – cannot visit a city more than once
- $\tilde{S} \subseteq S$ – all permutations of cities (with respect to Ω)
- Find globally minimum tour



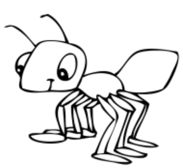
NP-hard combinatorial problems

- Cannot be **exactly** solved in polynomial time
- **Approximate** methods – generate near-optimal solutions in reasonable time
- No formal theoretical guarantees
- Approximate methods = **heuristics**



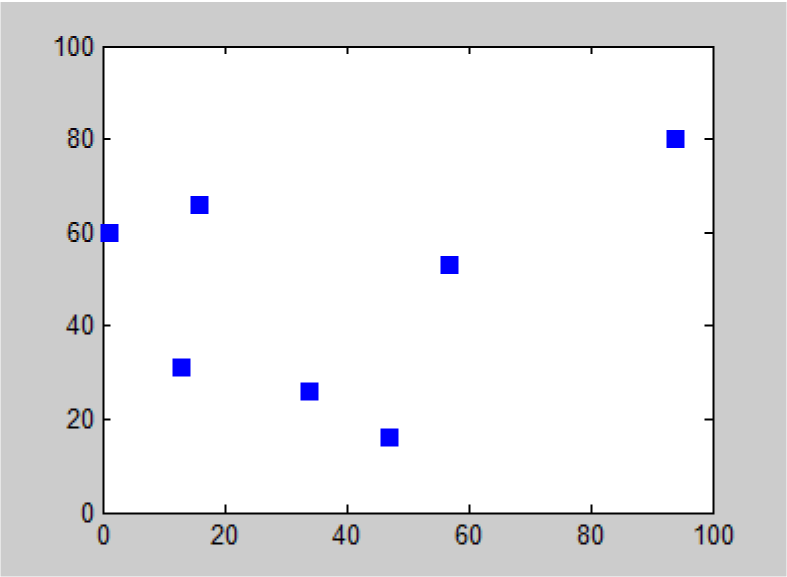
Approximate methods

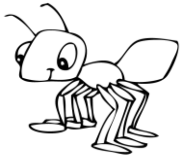
- Constructive algorithms
- Local search



Constructive algorithms

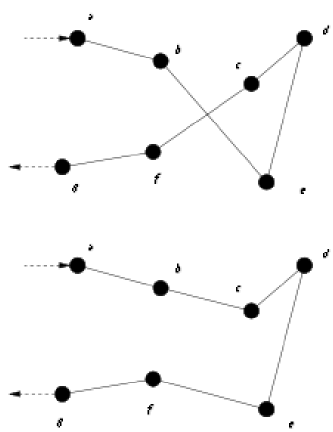
- Add components to solution incrementally
- Example – greedy heuristics: Nearest-Neighbour





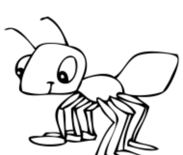
Local search

- Explore neighborhoods of complete solutions
- Locally optimize them
- Example: 2opt algorithm



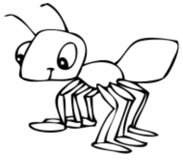
```
procedure 2optSwap(route, v1, v2) {
1. take route[0] to route[v1] and add them in
   order to new_route
2. take route[v1+1] to route[v2] and add them
   in reverse order to new_route
3. take route[v2+1] to route[start] and add
   them in order to new_route
4. return new_route; }
```

- Example route: A → B → E → D → C → F → G → H → A
- Example parameters: v1=1, v2=4 (assuming starting index is 0)
- Contents of new_route by step:
  - **(A → B)**
  - A → B → **(C → D → E)**
  - A → B → C → D → E → **(F → G → H → A)**

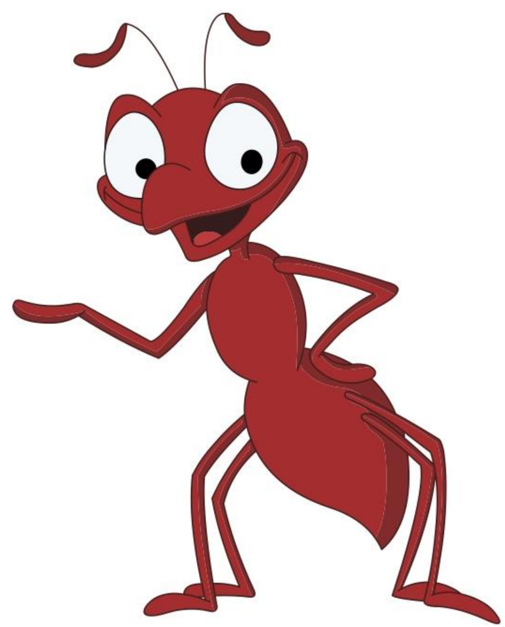


What is a metaheuristic?

- Heuristic function is parameterized
- High-level algorithm is designed to find best parameters of the heuristic function
- Applicable to a wide set of problems
- Example algorithms:
    - Simulated annealing
    - Tabu search
    - Iterated local search
    - Evolutionary computation
    - **Ant colony optimization**
    - Particle swarm optimization
    - etc.



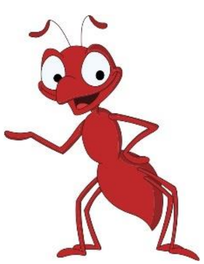
ACO for the Traveling Salesman Problem



Traveling salesman problem

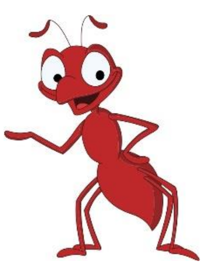
- N – set of nodes (cities), $|N| = n$
- A – set of arcs, fully connecting N
- Weighted graph $G = (N, A)$
- Each arc has a weight d_{ij} – distance
- Problem:

Find minimum length tour



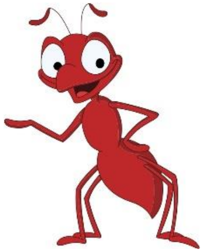
TSP: constraints

- All cities have to be visited
- Each city – only once
- Enforcing – allow ants only to go to new nodes



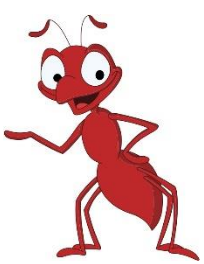
TSP: heuristics

- Pheromone trails - τ_{ij} :
  - Ants deposit pheromone trails as they travel from i to j
  - Pheromones affect the desirability of visiting city *j* directly after *i*
  - Pheromone initialization to constant: $\tau_{ij} = 1$
- Visibility - η_{ij} :
  - The longer the distance between i to j, the less visible it is for the ant
  - Visibility affects the desirability of visiting city j after I
  - $\eta_{ij} = 1 / d_{ij}$



TSP: solution construction

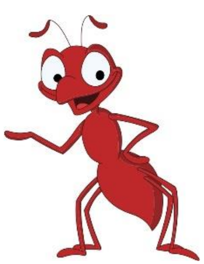
- Select random start city
- Add unvisited cities iteratively
- Until a complete tour is built



Ant System: Tour construction

- Ant *k* is located in city *i*
- N_i^k is the neighborhood of city *i*
- Probability to go to city $j \in N_i^k$:

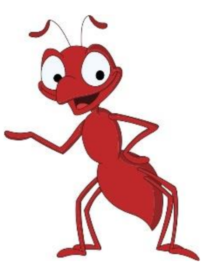
$$p_{ij}^k = \frac{\tau_{ij}^{\alpha} \cdot \eta_{ij}^{\beta}}{\sum_{l \in N_i^k} \tau_{il}^{\alpha} \cdot \eta_{il}^{\beta}}$$



Tour construction: comprehension

$$p_{ij}^{k} = \frac{\tau_{ij}^{\alpha} \cdot \eta_{ij}^{\beta}}{\sum_{l \in N_i^k} \tau_{il}^{\alpha} \cdot \eta_{il}^{\beta}}$$

- $\alpha = 0$ – greedy algorithm
- $\beta = 0$ – only pheromone is at work
  - quickly leads to stagnation



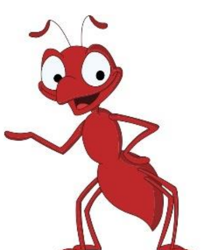
Ant System:
update pheromone trails – evaporation

Evaporation for all connections $\forall(i, j) \in L$:

$$\tau_{ij} \leftarrow (1 - \rho)\, \tau_{ij},$$

$\rho \in [0, 1]$ – evaporation rate

Prevents convergence to suboptimal solutions

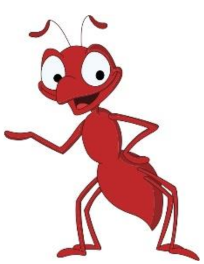


Ant System: update pheromone trails – deposit

- T^k – path of ant k
- C^k – *length of path* T^k
- Ants deposit pheromone on visited arcs:

$$\tau_{ij} \leftarrow \tau_{ij} + \sum_{k=1}^{m} \Delta\tau_{ij}^{k}, \forall (i, j) \in L$$

$$\Delta\tau_{ij}^{k} = \begin{cases} 1/C^k, (i, j) \in T^k \\ 0, otherwise \end{cases}$$

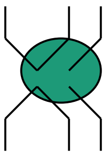


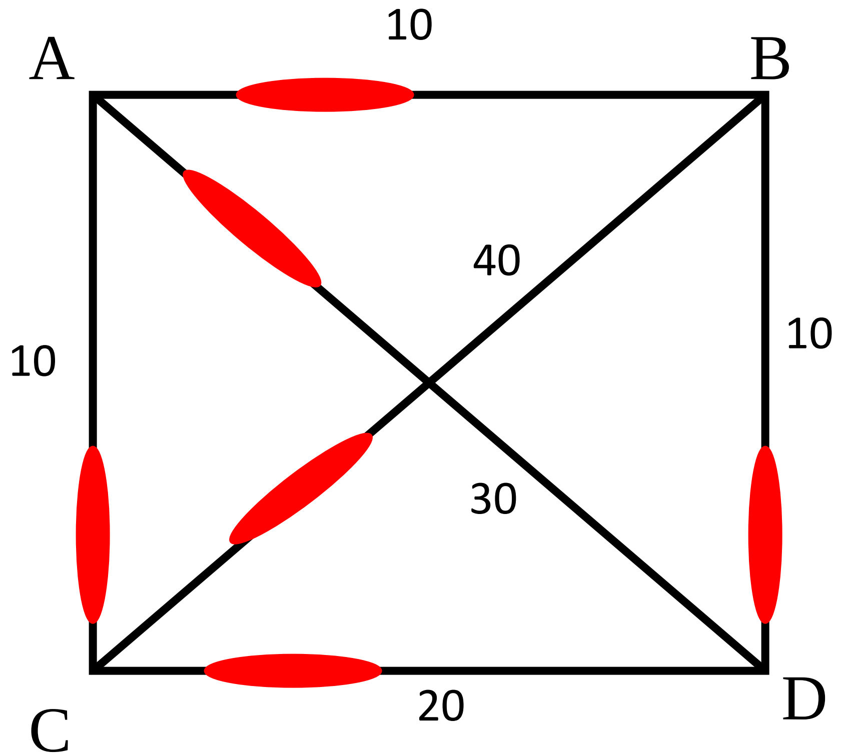
Ant Colony Optimization Algorithm

```
procedure ACO
    begin
        initialize the pheromone
        while stopping criterion not satisfied do
            position each ant on a starting node
            repeat
                    for each ant do
                            chose next node
                    end for
            until every ant has build a solution
            update the pheromone
        end while
    end
```

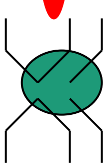
E.g. A 4-city TSP

Initially, random levels of pheromone are scattered on the edges





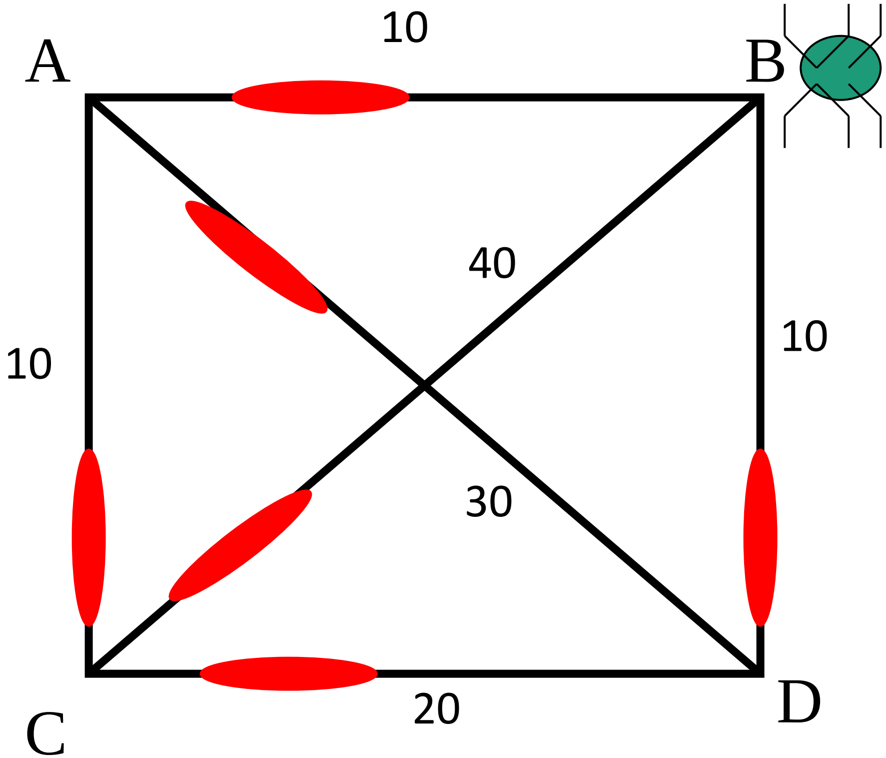
Pheromone



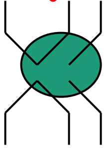
Ant

E.g. A 4-city TSP

An ant is placed at a random node



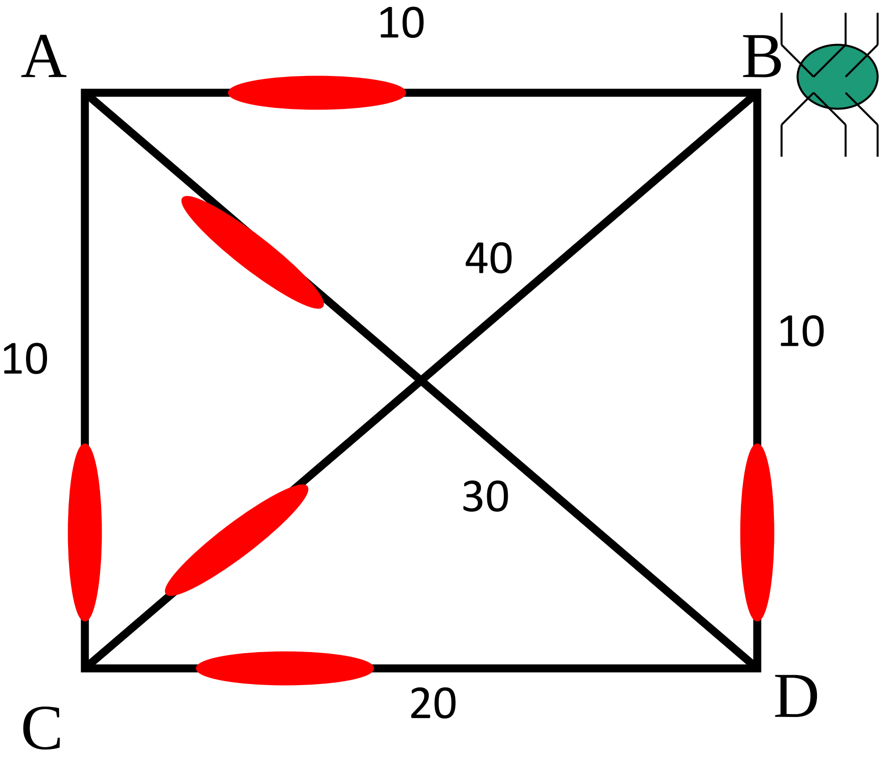

Pheromone


Ant

E.g. A 4-city TSP

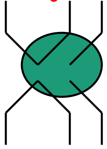
The ant decides where to go from that node, based on probabilities calculated from:

- pheromone strengths,
- next-hop distances.



Suppose this one chooses BC

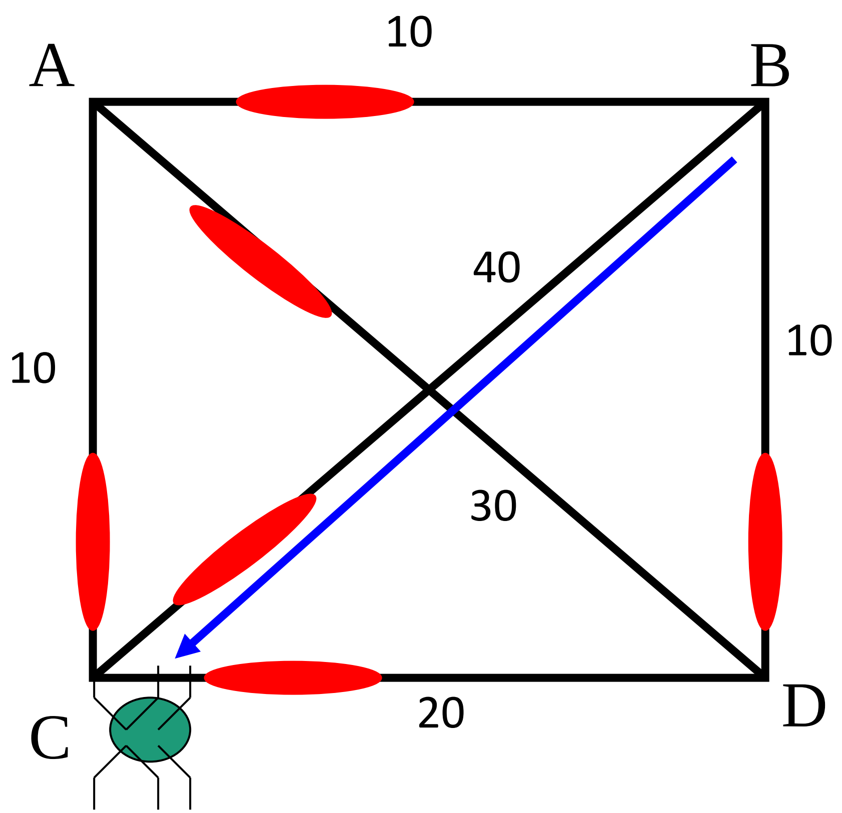
 Pheromone

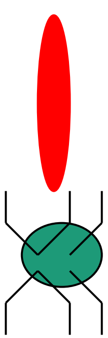
 Ant

E.g. A 4-city TSP

The ant is now at C, and has a `tour memory' = {B, C} – so he cannot visit B or C again.

Again, he decides next hop (from those allowed) based on pheromone strength and distance; suppose he chooses CD



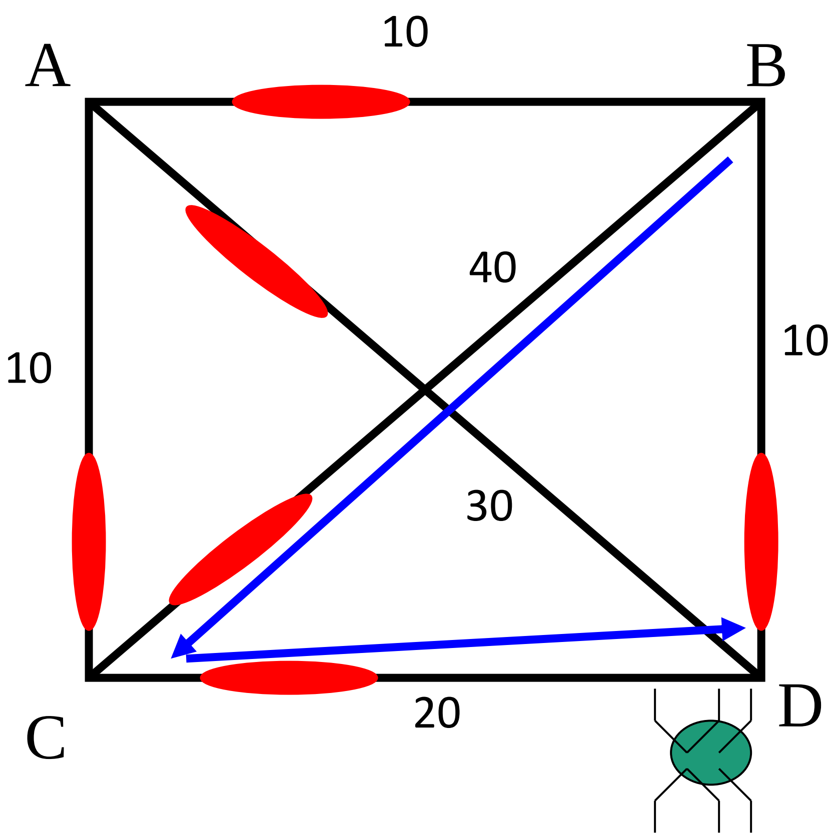


Pheromone

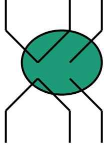
Ant

E.g. A 4-city TSP

The ant is now at D, and has a `tour memory' = {B, C, D}
There is only one place he can go now:

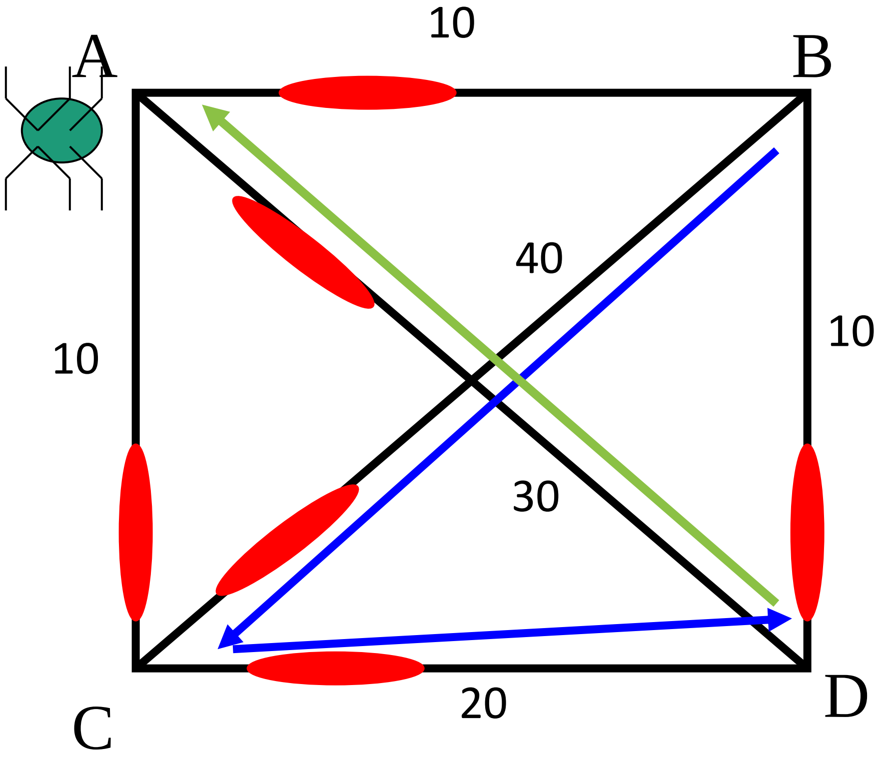


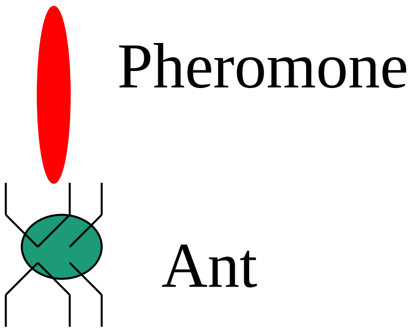

Pheromone


Ant

E.g. A 4-city TSP

So, he has nearly finished his tour, having gone over the links: BC, CD, and DA.

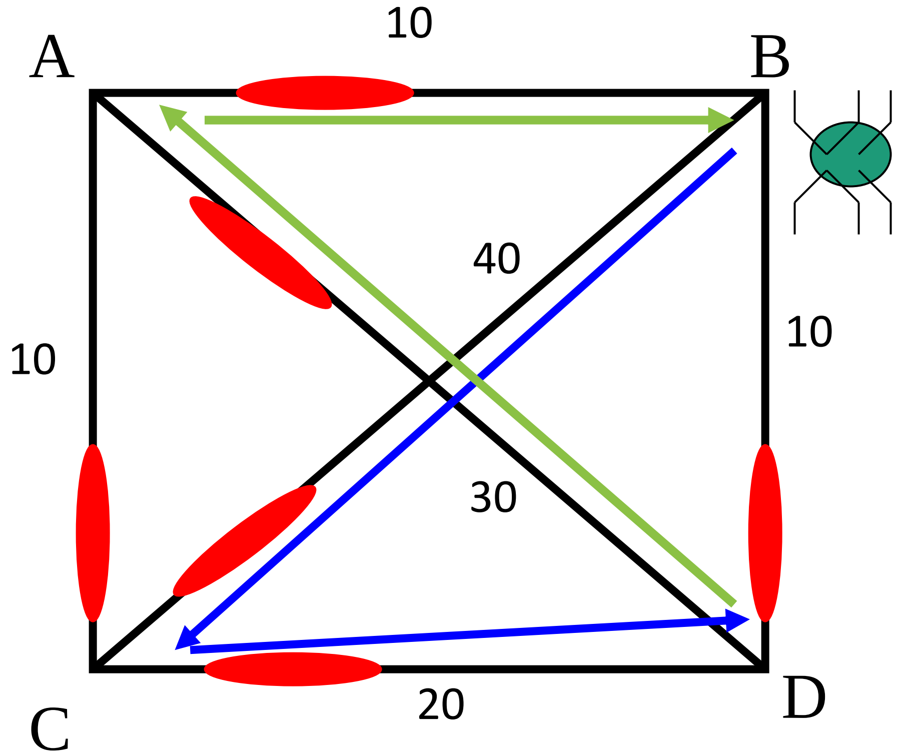


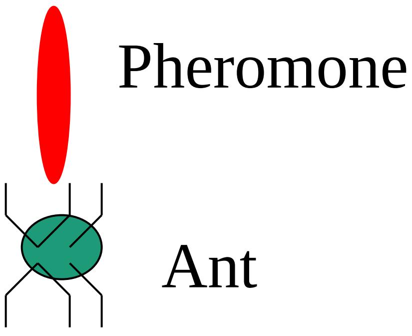


E.g. A 4-city TSP

So, he has nearly finished his tour, having gone over the links: BC, CD, and DA. AB is added to complete the round trip.

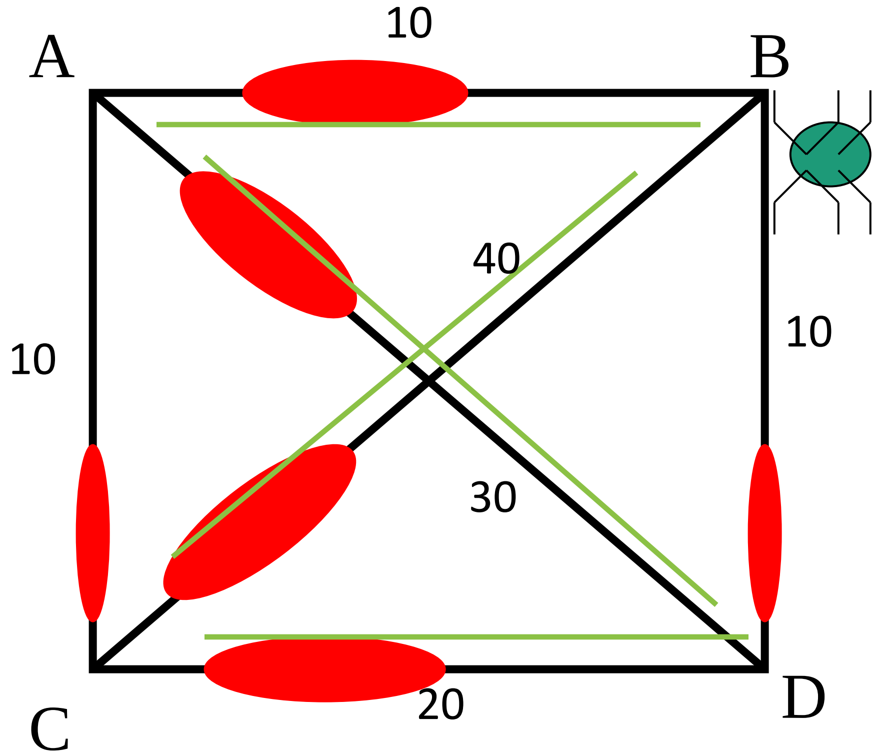
Now, pheromone on the tour
is increased, in line with the
fitness of that tour.

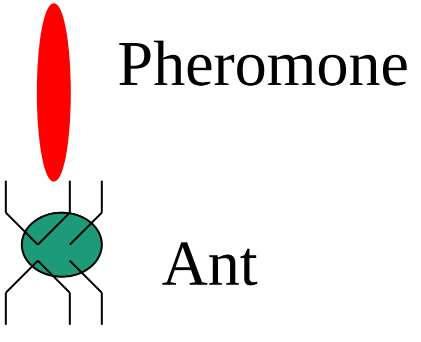




E.g. A 4-city TSP

Next, pheromone everywhere is decreased a little, to model decay of trail strength over time



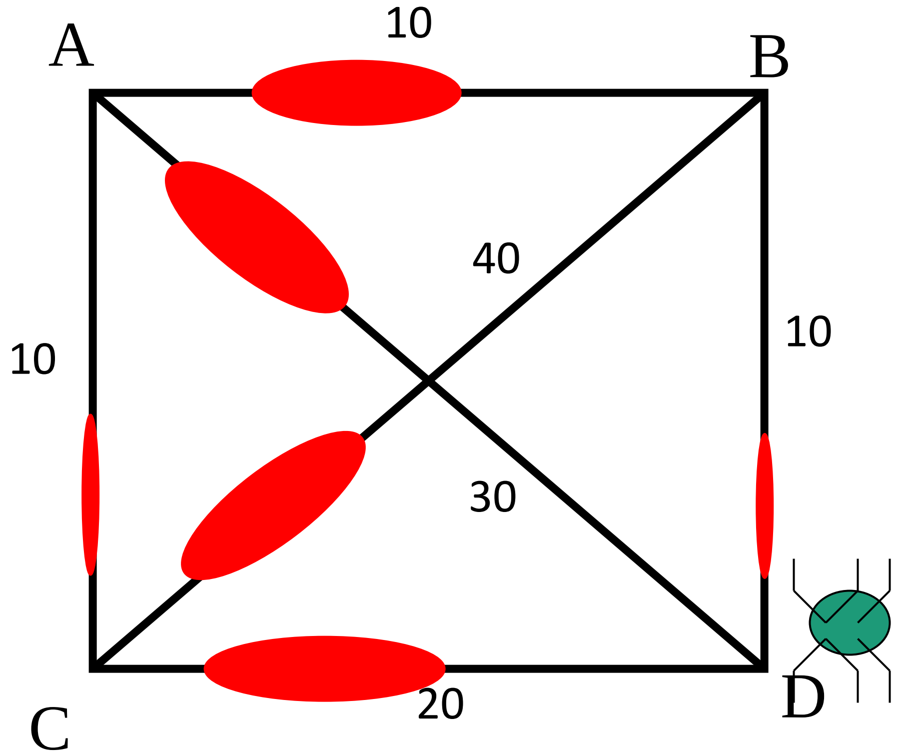


E.g. A 4-city TSP

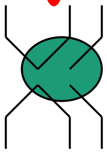
We start again, with another ant in a random position.

Where will he go?

Next, the actual algorithm and variants.



Pheromone



Ant

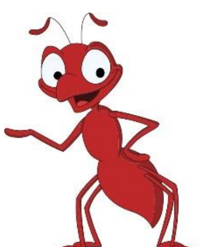
ACO applications

- Traveling salesman
- Quadratic assignment
- Graph coloring
- Multiple knapsack
- Set covering
- Maximum clique
- Bin packing

..

State of the art in TSP

- CONCORDE
  http://www.tsp.gatech.edu/concorde.html
- Solved an instance of 85900 cities
- Computation took 286-2719 CPU days!



Further reading

- M. Dorigo, T. Stützle. Ant Colony Optimization. MIT Press, 2004.
- http://iridia.ulb.ac.be/~mdorigo/ACO/